# Railway Signalling for Technicians

An introduction to principles, equipment and systems

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# Start here

This book provides a starting point for learning about railway signalling. Begin with the Introduction for its purpose, intended readership and study guidance, then read the opening chapters in sequence.

Overview of a signalling scheme brings the main system relationships together. The equipment chapters develop those subjects, and Useful other sources of information provides references for further study.

> ! Authority for work
>
> This is an educational introduction. It does not specify test limits, maintenance intervals, operating rules or authority to work. Use the applicable controlled documentation, local arrangements and competence requirements for work on the railway.

# 1. Introduction

This book is intended to be used as a starting point for those learning about railway signalling for the first time. It is designed to give a basic overview and understanding of signalling, and an introduction to the railway.

It is essentially my attempt at creating the book I wished I had when I first started.

## 1.1 Who the book is for

The book is intended for those who are new to railway signalling, including those beginning work in signalling and telecommunications (S&T). It introduces the railway equipment, its purpose and the relationships between systems before moving into more detailed equipment descriptions.

## 1.2 How to use the book

Read the opening chapters in sequence. Railway basics introduces line names, directions and location references. Railway signalling introduces train separation and control, while Overview of a signalling scheme brings the main equipment and system relationships together. The later chapters develop trackwork, points, signals, circuit reading, relays and the systems that connect them.

For each item, establish what it does, how it fits into the wider signalling system and what happens when it fails to perform its function.

Use Useful other sources of information to find references for further study. Practical work requires the applicable controlled documentation, local arrangements and competence; this introductory book does not define test limits, maintenance intervals or authority to work.

# 2. Railway basics

**i** Learning objectives

Understand common terminology used to refer to trackside locations, understand directional terminology on the railway, and how to identify locations.

## 2.1 Responsibilities

Railway lines are formed of two running rails and sleepers between them. There are also various other components that you will see around the railway, some of which will belong to other departments. The track department, more commonly known as p-way, is responsible for looking after the track itself. This includes the rails, the sleepers, and the ballast.

**Background**

The name "p-way" comes from the term "Permanent Way", which was used to describe the final railway that was constructed. In the early days of railway building a "Temporary Way" was first quickly constructed to help get materials to site.

Almost all of the railway surrounding Glasgow is now electrified using overhead lines. There are therefore overhead gantries and stanchions dotted around to suspend the overhead lines from. The Overhead line department are responsible for looking after these, although the power is maintained by the Electrification and Plant (E&P) department. E&P are also responsible for switch heaters that run along the rails of points to keep them clear of snow and ice, and the compressed air which drives many of the points in Glasgow Central.

Off-track look after all the things which are not directly part of the railway, but are needed to keep it safe and secure. This includes fencing and vegetation. You might need to speak with them a lot in the summer months when trees begin to grow and obscure signals.

The Telecoms department are responsible for all the telecoms that is used by the railway. In years gone by this was part of the signalling department's job, which is why we're known as S&T, although now we do very little telecoms. The only telecoms we are still responsible for are signal post telephones (SPTs).

## 2.2 Tracks and lines

A **track** normally consists of two running rails and the structure that supports them. A **single-track** railway has one track; a **double-track** railway has two. A railway with four tracks has eight running rails. Points connect tracks so that trains can pass from one to another.

The word **line** often means a particular track, as in *Up Main line*. It can also describe a railway between places, so someone might say the Neilston line, but actually mean both the *Up Neilston* and the *Down Neilston* as they're referring to the branch as a whole. A bit of context is often needed to know exactly which is being referred to.

## 2.3 Up and Down directions

**Up** and **Down** are established railway directions. They provide a consistent way to describe movements and distinguish tracks without relying on the direction in which someone happens to be looking. Up commonly means towards the principal station on that route, often the headquarters of the company which built that line. Many lines use London as the marker for the Up direction, but some lines in the area use Glasgow instead. This is just something that you will have to learn as you get to know the section.

Most lines will have one predominant direction of movement, so will be known as the Up or Down line. For example, the *Up Fast* mostly carries trains in the Up direction, towards London. Some bits of it are bi-directionally signalled however, so trains can travel in either direction. This means it is possible for a train to be travelling in the Down direction (towards Glasgow) on the *Up Fast*.

Where there is a single-track section of railway these are normally bi-directionally signalled so that trains can travel in either direction. They will still have an Up and a Down direction though.

## 2.4 Line names

Fast and Slow are common line names, although in some regions the terms Main and Relief are more commonly used, with the same meanings. Fast lines are generally intended for faster or non-stopping services, with Slow lines more commonly associated with stopping passenger services or freight trains. Fast lines might have a higher maximum permissible speed than Slow lines, but this isn't always the case. Often the line speed might actually be the same for both lines.

Lines may also have other designations, which might refer to geographical locations (e.g. Gourock or Hertford), usage (e.g. Suburban or Departure), or be named for historical reasons (e.g. City Union or Smithy Lye). These designations are not necessarily an indication of how a line is used today, and their meaning may have been all but forgotten.

## 2.5 ELRs, Track IDs, and mileages

Every location on the railway has an ELR, Track ID, and mileage. With these three pieces of information every location on the railway can be identified exactly.

### 2.5.1 Engineer's Line Reference

Each route on the railway is given an Engineer's Line Reference (ELR). This a three letter code, which sometimes also has a number. ELRs were generally chosen wherever possible to relate to the line that they referred to, but as each line needed a unique ELR this wasn't always possible.

A number is used at the end of the ELR when there is a discontinuity in mileage along a route. This might happen if the track layout has changed over the years or the original measurements were very inaccurate.

The ELRs within the Glasgow Central section are given in the table below. Origins of the ELR are given where they aren't obvious to help remember them.

| ELR | Line | Name Origin |
|---|---|---|
| ARG1 & ARG2 | Argyle line | |
| AYR1 | Ayr line | |
| BRD | Through Terminus | This line is also known as the **B**urma **R**oa**d** |
| CLY | Clydesdale line | |
| CNC | Cathcart North Curve | |
| CND1 & 2 | Deanside branch | The line branches off at **C**ardo**n**al**d** |
| CNL | Corkerhill and Paisley Canal line | |
| CTC | Cathcart Circle | |
| EKE | East Kilbride line | |
| GBK | Barrhead line | Originally built by the **G**lasgow, **B**arrhead and **K**ilmarnock Joint Railway |
| HST | City Union line | Line joins Shields Junction and **H**igh **St**reet |
| KHL | Kirkhill line | |

| ELR | Line | Name Origin |
|---|---|---|
| LFS | Larkfield Curve | Originally known as the **L**ark**f**ield **S**outh Curve |
| LYE | Smithy Lye | |
| MEN1 & MEN2 | Muirhouse line | Probably named as the line is from **M**uirhouse Junction to **E**glinto**n** Junction |
| RCB | Carmyle line | Originally known as the **R**utherglen and **C**armyle **B**ranch |
| RNC | Rutherglen West Curve | May have once been known as the **R**utherglen **N**orth **C**urve |
| TSS | Terminus Curve | Originally known as the **T**erminu**s** **S**outh Curve |
| WCM2 | West Coast Mainline | |

### 2.5.2 Track ID

Each line also has a Track ID, which is a four digit code which describes the type of line and the direction of travel.

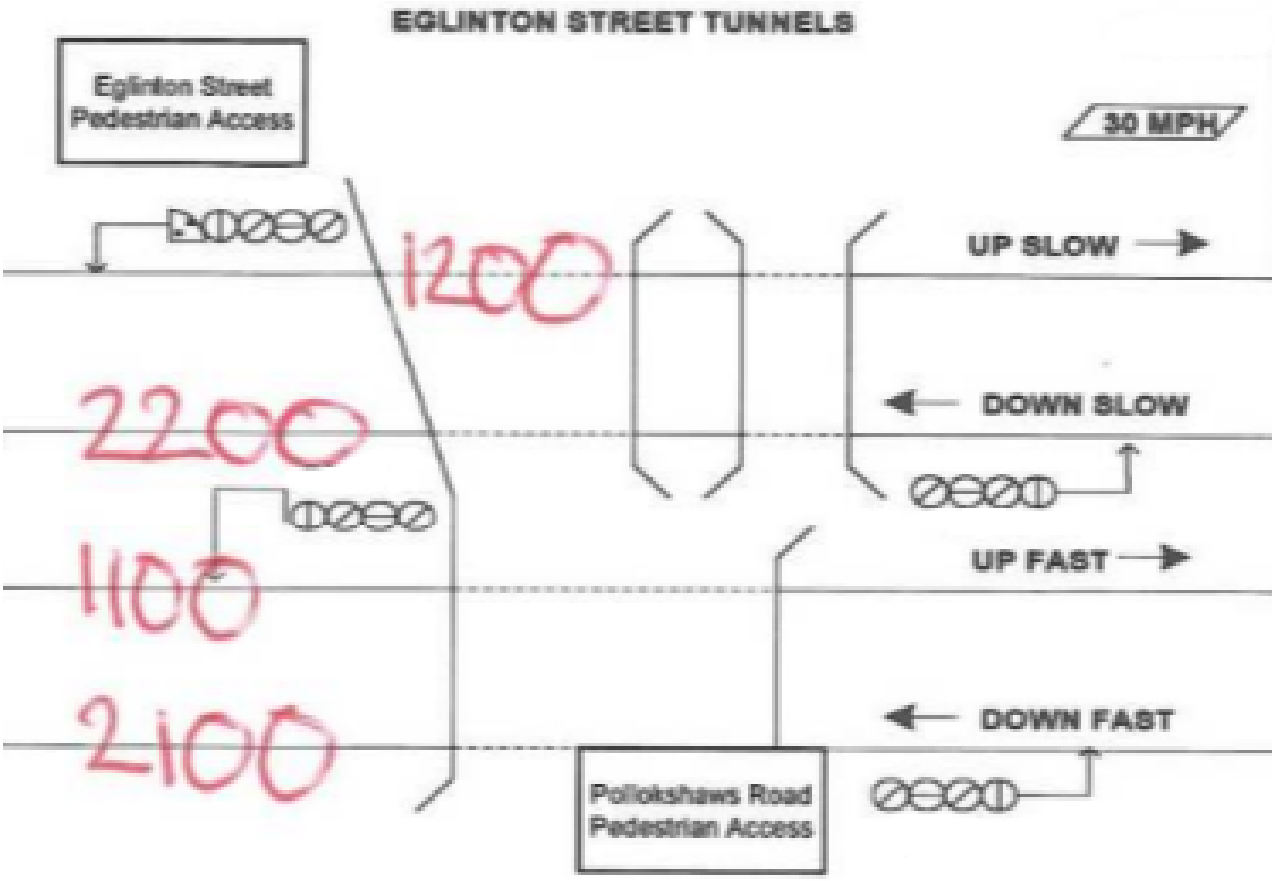
Track IDs aren't used often by S&T, so it isn't strictly necessary to memorise them all but they can be helpful when talking to other departments which use them more often, such as pway.

The numbers are always in the form of DNXX where:

| D | N | XX |
|---|---|---|
| 1 - Up | 1 - Main or Fast (or other line performing this function) | 00 - the first line at this location with that direction and type |
| 2 - Down | 2 - Relief, Slow, or Through (or other line performing this function) | 01 - the second line at this location with that direction and type |
| 3 - Bi-directional | 3 - Goods | 02 - the third line at this location with that direction and type |
| 4 - Merry-go-round loop | 4 - Single line | ...and so on... |
| | 5 - Loop | |
| | 6 - Terminal track or Bay | |
| | 7 - Crossover | |
| | 8 - Other | |
| | 9 - Siding | |

Using the below diagram from Glasgow Central as an example we have:

- 1200 - Up Slow
- 2200 - Down Slow
- 1100 - Up Fast
- 2100 - Down Fast




### 2.5.3 Mileage

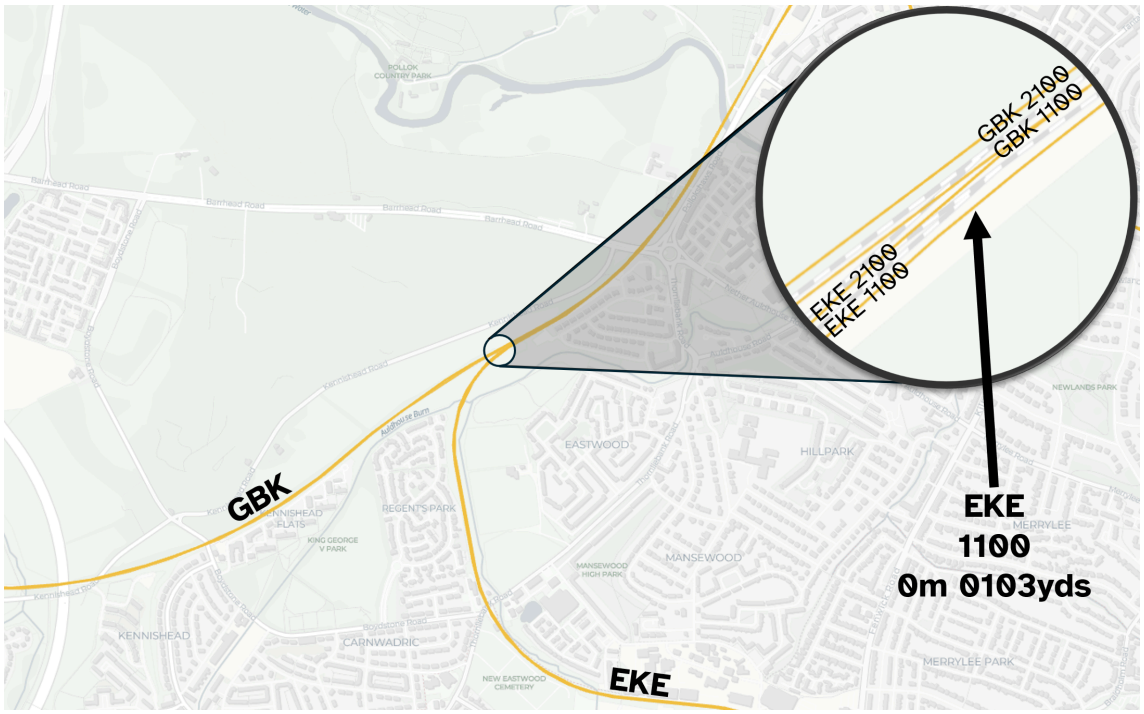
All lines also have a mileage, starting from a datum point. The origin points are normally the destination of the Up line, but there are some exceptions.
In Scotland mileages are usually given in Miles and Yards, but in the rest of the UK mileages are given in Miles and Chain. You might still see chain used in some documents, such as the Sectional Appendix, so it's important to be be able to convert between the two.

A Chain is an old measurement equal to 22 yards (or roughly 20 metres).
1 mile = 80 chain = 1760 yards
3/4 mile = 60 chain = 1320 yards
1/2 mile = 40 chain = 880 yards
1/4 mile = 20 chain = 440 yards

Mileposts are used to mark every quarter of a mile (although they are sometimes missing, or hidden). The large number is the mile, and the bars or dots underneath indicate quarters of a mile.



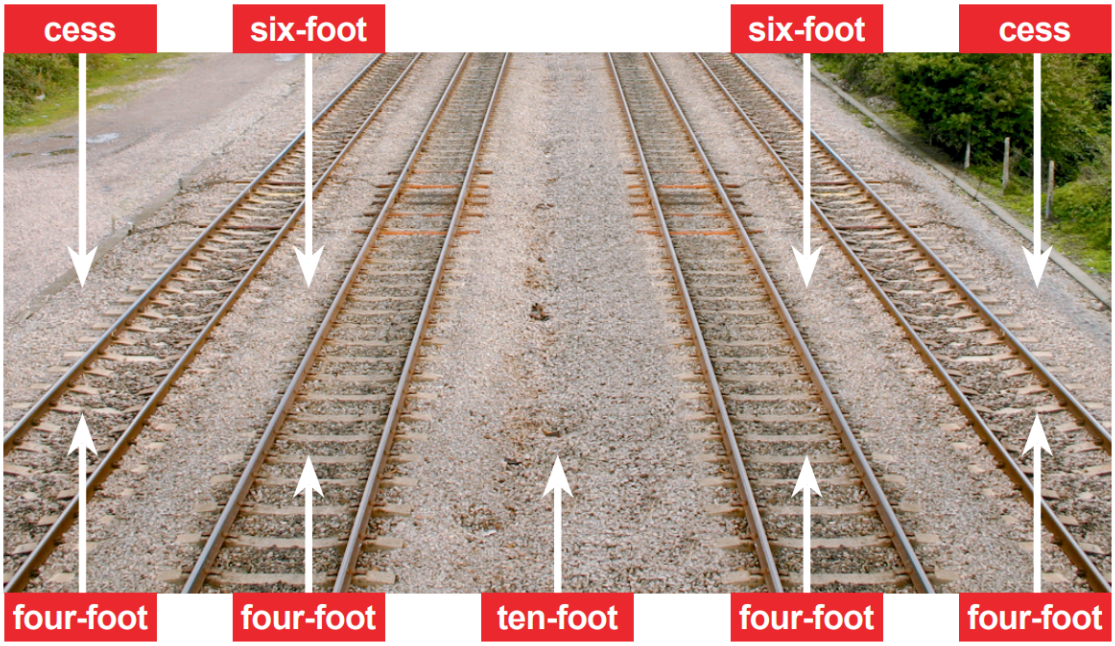
The diagram below of Busby Junction shows how a location can be narrowed down first by ELR, then by Track ID, and then by mileage. Alternatively, just the ELR and mileage could be used to identify a place but not necessarily a track, or a line name could be used. For this example it would be just as valid to say 0m 103yds on the *Up East Kilbride*.




## 2.6 The spaces around the track

The following names describe where something lies relative to the rails:

| Term | Meaning |
|---|---|
| **Four-foot** | The space between the running rails of one track |
| **Six-foot** | The space between adjacent tracks in the usual double-track description |
| **Ten-foot** | The wider space between pairs of tracks in a four-track arrangement |
| **Cess** | The area alongside the outside of the running lines |




These names identify spaces rather than certifying their dimensions, for example, *four-foot* is not the measurement used to specify track gauge. The Trackwork chapter explains gauge and the construction of the track.

You may also hear the 10-ft called a “wideway”.

> ! Important
>
> Spaces may be smaller than their names suggest. Do not rely on the names to judge whether you are in a position of safety.

## 2.7 How tracks appear on plans

A signalling scheme M12 plan generally represents each track by **one line**. That line stands for the complete track, not an individual rail. A double-track railway therefore appears as two lines even though it has four running rails. A detailed bonding diagram, by contrast, generally shows both rails as the individual connections between the rails matter.

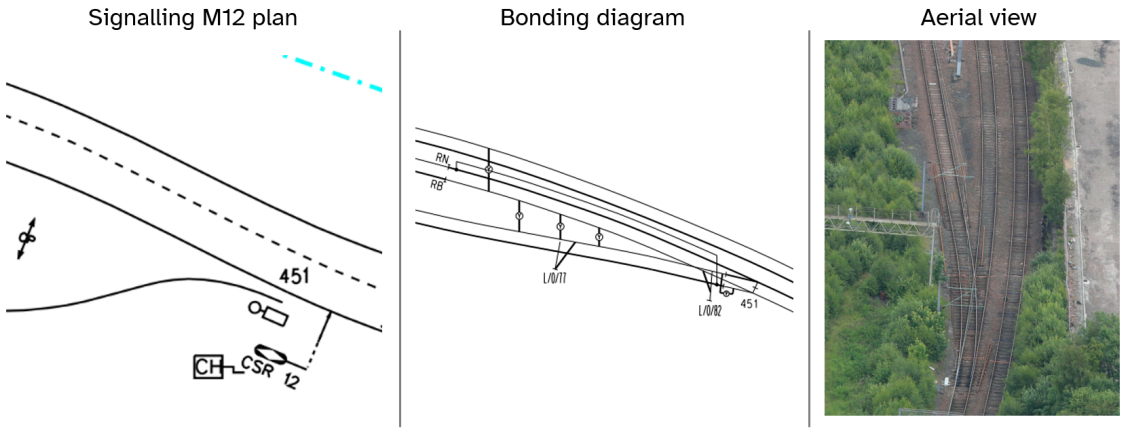



Figure 2.1: Same junction on both an M12 signalling plan, a bonding diagram, and in real life

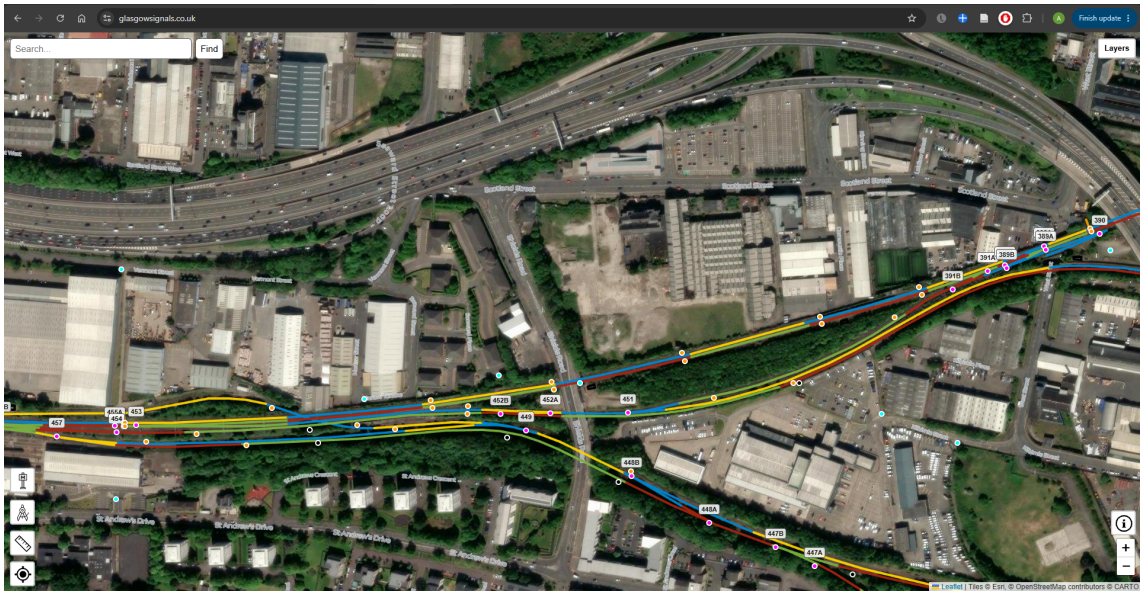
M12 diagrams are the most common diagram that you will come across. They show almost all signalling equipment, but can be confusing at first due to the number of symbols used. The above diagram contains the symbol for a point machine (the connected circle and rectangle).

> **Tip**
>
> Start by finding the orientation of a plan, using line names, stations, points, signals, or other features that are shown. It will become easier to relate what you see on the plan to what you see on the ground as you become more familiar with both the plans and the location.

It's important to understand what is shown on each type of diagram so that you can relate it to the real world conditions on the ground.

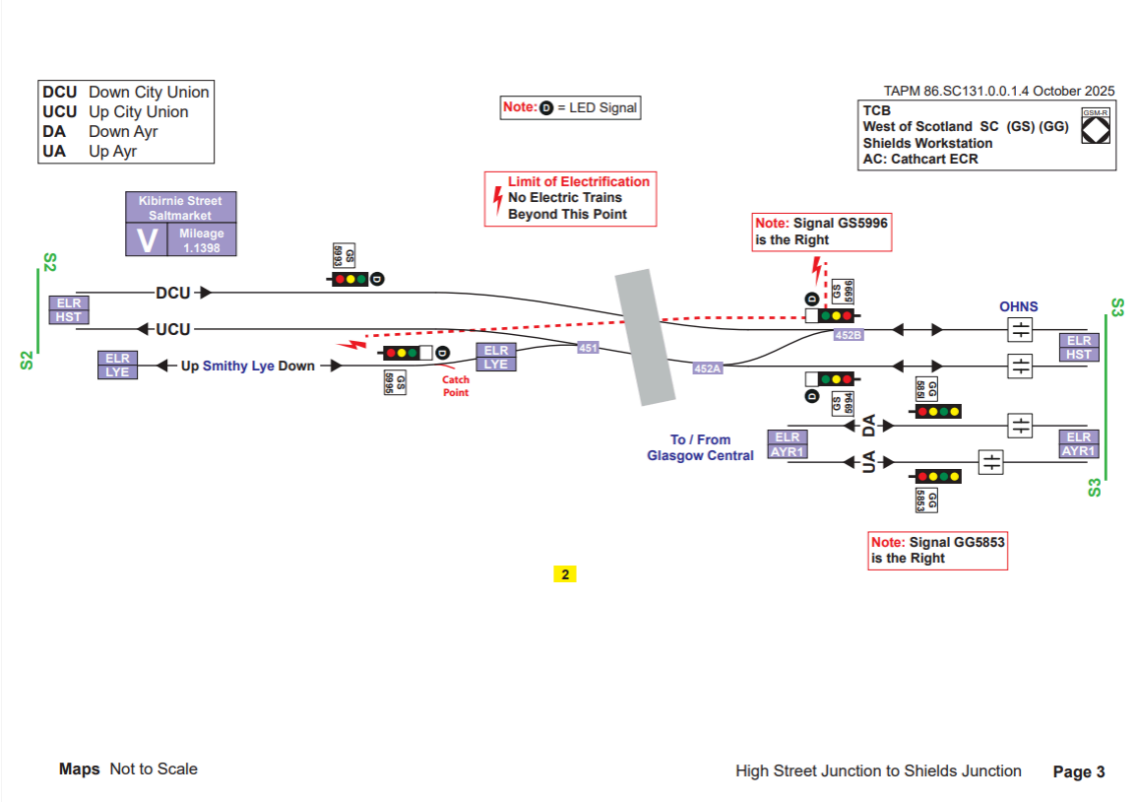
M12 signalling plans and bonding diagrams are generally to scale, but the scale may change throughout the plan depending on the complexity of the layout. Do not assume that just because two things look close together on the plan they will be a short walk in real life!

There is also an unofficial website [password: Glasg0w!] covering Glasgow showing the locations of all the assets and track circuits which can be useful when working out distances or finding assets. It's vital to be able to understand official diagrams but this can be useful resource to assist you in getting to know the area.




## 2.8 Other diagrams

Another diagram you may come across are TAP diagrams. These can be found on the Scotland's Railway Planning Hub. They can be a bit more user friendly, but aren't used that often as they don't give as much detail as an M12.




The Sectional Appendix is a document produced that gives all sorts of information about each line, which is mostly relevant to train drivers. It does, however, include diagrams which contain some important information, such as line speeds and controlling signal box. The relevant extracts of the Sectional Appendix will always be included in your safe work pack and can also be found online.

| LOR | Seq. | Line of Route Description | ELR | Route | Last Updated |
|---|---|---|---|---|---|
| SC131 | 001 | High Street Jn to Shields Jn | HST | Scotland | 20/02/2016 |

| Location | Mileage M Ch | Running lines & speed restrictions | Signalling & Remarks |
|---|---|---|---|
| High Street Jn | 0 04 | SC123 seq 5 | TCB Yoker SC (YF) East Workstation |
| | | To Cook St SC059 seq 3 | West of Scotland SC (G) Shields Workstation |
| OHNS | 2 16 | | |
| | 2 17 * | | |
| Shields Jn | 2 35 | SC059 seq 3 / To Shields Jn SC029 seq 001 | |

# 3. Railway signalling

**i** Learning objectives

Explain the need for train separation, distinguish block working from interlocking, and describe the relationship between electrical operation, train detection and route setting.

## 3.1 Safety and the service

The principal purpose of signalling is to organise movements so that trains do not collide. That includes a following train catching a train ahead, opposing trains meeting on the same line, and trains converging at a junction. Signalling also influences the service that can be operated. A railway carrying occasional trains has different capacity requirements from one carrying trains a few minutes apart.

These purposes are connected. Shortening the interval between trains requires a suitable method of separation and adequate warning for braking. Adding signals without considering braking, detection and route locking does not automatically create a useful increase in capacity.

## 3.2 Why time alone is insufficient

A time-interval system releases a second train a specified time after the first. Its weakness is simple: the first train may have stopped. Elapsed time cannot prove that the line ahead is clear.

A space-interval system instead separates trains by defined sections. The key question becomes whether the preceding train has left the protected section. This is the foundation of block working. How the railway establishes that a section is available depends on the system and its associated operating arrangements.

## 3.3 Absolute block

In a simple example, adjacent signal boxes A and B communicate using bells and block instruments. The receiving signaller controls indications associated with the section. The three indications are **Line Blocked**, **Line Clear** and **Train on Line**.

At a conceptual level, the receiving box agrees to accept a train, the sending box signals it forward under the required conditions, and the instruments record its entry into the section. The receiving box establishes that the train has arrived complete before the section is made available again. Arrival of the locomotive alone is insufficient: vehicles could have become detached and remain in the section.

This explanation identifies the safety purpose of the communications. It is not a bell-code sequence or an instruction for operating block instruments.

**!** Two different protections

Block working organises the separation of trains along a line. Interlocking prevents incompatible settings and movements within its designed scope. A junction can be correctly set while the line ahead is occupied; an empty line does not prove that a junction is correctly set.

## 3.4 Electrical operation and control

A point machine provides the force at the points, while electrical detection reports the required resulting conditions. Relay circuits or electronic interlockings enforce the movement constraints, allowing the control interface and the equipment implementing the safety logic to have different physical forms. The operator's command, the equipment action and the resulting indication are separate parts of the system.

## 3.5 Track circuit block and automatic signals

A track circuit uses the rails as part of an electrical circuit to establish whether a defined section is clear. It gives the signalling system information that does not depend on the signaller remembering the position of every train.

With continuous train detection and suitable controls, a plain-line signal can respond automatically to conditions ahead. In track circuit block, train detection supports the separation of trains between signals. **Automatic** describes how a signal is controlled; it does not mean that it operates without safety conditions.

The signal aspects must give the driver the required warning of a stop indication. The lineside signals chapter connects that requirement to braking distance, overlaps and headway.

## 3.6 From individual controls to route setting

With individual controls, the signaller sets the points and then operates the signal control. An entrance-exit (NX) interface lets the signaller identify the start and end of the required movement. The system calls the required points and establishes the route conditions.

The route command does not override the interlocking. If a required point is not detected, a track section is occupied or a conflicting route remains locked, the commanded proceed aspect cannot be obtained under those conditions.

Remote control makes it possible to operate geographically separated interlockings from a signalling centre without providing a separate long-distance wire for every control and indication. The interlocking and remote control system perform different jobs, a distinction developed later in the course.

## 3.7 Train describers: background

A **train describer** associates a service identity with a displayed position, or berth, and uses train-detection and route information to step the description as the train moves. Its information can also support passenger displays and other operational systems. It identifies the service associated with a position; it does not prove the track clear or replace the interlocking's safety checks. A wrong description can therefore cause confusion or misrouting even when the interlocking continues to prevent conflicting movements.

## 3.8 Automatic Warning System

The Automatic Warning System (AWS) provides an audible indication in the cab. A warning requires acknowledgement by the driver; if it is not acknowledged, the system initiates a brake application.

AWS supports the driver by drawing attention to a warning. It works alongside the signals and the driver's knowledge of the route.

## 3.9 Activity: two trains at a junction

Train A is standing beyond a junction. Train B is approaching a signal before it. The points are correctly set for B's intended route. Explain why that fact alone cannot justify clearing B's signal.

> **i** Suggested reasoning
>
> Correct point position establishes only one condition. The line to be used must also satisfy the train-detection and separation requirements, the required overlap and protection must be available, and conflicting movements must be excluded. A train standing beyond the junction may prevent these conditions being met.

## 3.10 Key points

- Time elapsed does not prove that a train has cleared a section.
- Block working and interlocking address related but distinct problems.
- Route setting simplifies entering commands; the interlocking still governs the result.

- AWS draws attention to a warning and requires the driver to acknowledge it.

# 4. Overview of a signalling scheme

**i** Learning objectives

Identify the main parts of a signalling system; follow a movement through command, action and indication; explain the roles of power, communications and equipment housings; and distinguish right-side from wrong-side failures.

## 4.1 The system as a whole

A railway moves vehicles along a guided path. A train cannot steer round an obstruction and may need a substantial distance to stop. Signalling organises movements so that the required route and separation are established before the train proceeds.

Track provides the path, points select it, train detection reports occupation, and the interlocking determines when a movement may be authorised. Signals communicate the permitted movement to the driver. Power supplies and telecommunications support those functions.

A useful first drawing separates the **control device**, **interlocking** and **railway equipment**. Commands travel towards the equipment; indications return information about its state.

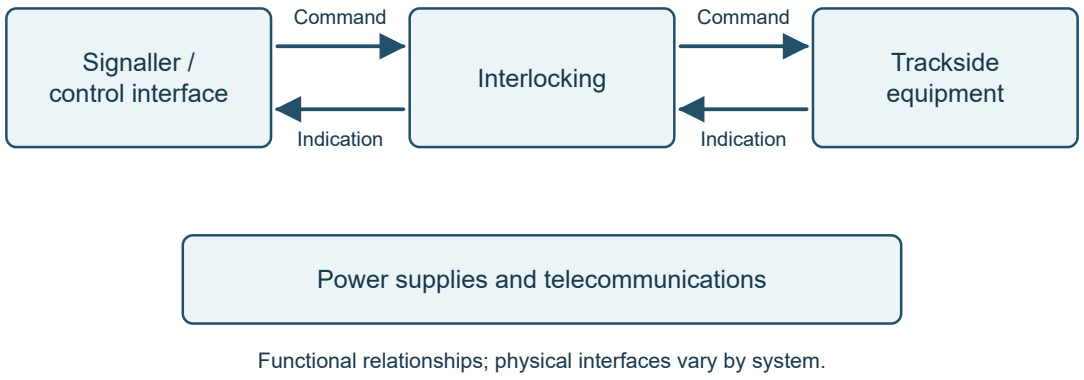



Figure 4.1: The principal control and indication relationships. Power and telecommunications support the system.

The arrows are functional relationships, not a statement that every installation uses a separate cable for each arrow. Relay interlockings and SSI implement the relationships differently.

## 4.2 Command, action and indication

Follow the **command**, **action** and **indication** separately. The command tells the equipment what to do; the action is what the equipment does; the indication reports its state. A command can be registered without the equipment completing its movement, so an indication must be understood through the equipment and circuit arrangement.

The signaller uses a control panel or workstation to enter commands and view indications. A track layout and equipment indications may share the same screen, but their physical combination does not make their engineering functions identical.

The interlocking checks the designed conditions and controls the railway equipment accordingly. It receives information such as point detection, train-detection states and signal proving. The signaller's command passes through this logic before it can result in a permitted movement.

## 4.3 Follow a movement through the system

Imagine a train waiting at a signal before a junction. The signaller wants it to take the branch. Several distinct events must occur:

1. The signaller enters a route command.
2. The interlocking checks the conditions associated with that route.
3. Points are called to the required positions where movement is permitted.
4. Detection reports the positions and locking conditions required by the design.
5. The interlocking establishes that the necessary conditions are proved and that conflicting movements are excluded.
6. The signal can display the permitted aspect.
7. Train detection reports the train's progress, supporting the designed release of the route.

This sequence is simplified. Each stage depends on the actual equipment and controls: registering the command, supplying the operating energy, completing the action and obtaining the relevant indications are separate events.

## 4.4 Fail-safe behaviour

A **right-side failure** produces a more restrictive result than the traffic conditions require. For example, loss of a track-clear proving circuit may prevent a signal clearing even though the track is physically empty. A **wrong-side failure** produces a less restrictive result than the conditions justify, such as an occupied section being indicated clear to the safety logic.

Fail-safe design aims to prevent specified faults from producing unsafe permissions. It does not mean that equipment cannot fail, nor that every possible failure automatically lights a red lamp. A failed power supply can leave a signal dark. The safety arrangement must address the loss of the required information or indication as part of the design and operating system.

> ! Think in terms of permission
>
> The central question is whether the failure can allow a movement that should be prevented. "A fault has occurred" does not by itself tell you whether its effect is right-side or wrong-side.

## 4.5 Power is a separate engineering path

The logic deciding that points may move does not supply the mechanical energy needed to move them. Similarly, a command for a signal aspect does not itself supply the lamp's energy. Power supplies, distribution cables, transformers, rectifiers and local protection support the control system.

A signalling-centre power cubicle provides supplies at the required voltages, trackside distribution and standby arrangements. One arrangement uses 650 V distribution to trackside equipment, with local conversion to the required working supplies. Individual circuits in a location case operate at their specified voltages.

A common supply can affect several apparently unrelated items. When several indications disappear together, their shared power or communications path is a relevant system relationship to examine in the drawings.

## 4.6 Cable routes and equipment housings

Cables follow a planned route, often in troughing. Undertrack crossings carry routes between sides of the railway. The route protects and organises cables; it also brings the signalling system into contact with civil-engineering work, drainage and the track environment.

A **location case**, often called a **loc**, houses interfaces close to the equipment. It can contain terminals, relays, track-circuit apparatus, local supplies and other equipment appropriate to the installation. Short connections from a housing to the signal, point equipment or rails are commonly called **tail cables**.

An equipment room provides more space where a group of cases would be unsuitable. The important relationship is the same: equipment housings provide the interface between the wider scheme and local railway assets.

## 4.7 Automatic route setting and telecommunications

Automatic route setting can add a further layer of commands using timetable and train information. The commands remain subject to interlocking conditions.

Telecommunications provides both human communications and data paths. Signal post telephones, point-area telephones and radio are examples. A transmission network can also carry remote control and other operational information. The service being carried and the physical bearer carrying it should be considered separately.

## 4.8 Other interfaces

Level crossings and the Automatic Warning System (AWS) connect signalling with other parts of the railway. Each has a different purpose:

| Interface | Main purpose |
|---|---|
| Level crossing equipment | Manage the interface between road and rail movements |
| AWS | Provide an in-cab warning or clear indication associated with an approach |

These interfaces are introduced here to explain their place in the signalling system. Equipment-specific chapters can develop their operation in more detail.

## 4.9 Worked example: one symptom, several paths

A display shows that a branch route command has been registered and the points have been commanded reverse, but the entry signal remains at danger. Someone concludes that the signal must have failed because the command went through.

Registration of the command does not establish that the points moved, the required detection was obtained or every signal control condition was satisfied. Follow the system relationships:

1. **Command:** was the intended route or point command registered?
2. **Energy:** was the required supply available to the equipment?
3. **Action and indication:** did the equipment move and lock, and was the required detection indication obtained?
4. **Signal controls:** were all other conditions for clearing the signal satisfied?

A signal remaining at danger may be the correct response to a missing indication elsewhere. This example organises the evidence; it does not prescribe an order for intrusive tests or identify a failed component from the final symptom alone.

## 4.10 Key points

- Command, action and indication are distinct parts of the equipment sequence.
- A registered command does not establish that every subsequent stage has completed.
- The interlocking separates the movement command from the permission to act.
- Power, communications and physical railway equipment are all part of the scheme.
- Failure classification depends on the safety consequence.
- An interface drawing is a useful starting point for understanding a whole installation.

# 5. Trackwork

**i** Learning objectives

Describe the functions of rails, sleepers, fastenings and ballast; explain why joints and bonding matter to signalling; explain track gauge and the wheel–rail interface.

## 5.1 The track is part of the signalling environment

Track provides a load-bearing, guided path for the train. For S&T, it can also be part of an electrical circuit, a support for detection equipment and a mechanical reference for point operation. Changes to track geometry, rail joints or support can therefore affect signalling equipment.

A simple track assembly consists of two rails held at the required gauge by sleepers and fastenings. Ballast supports the sleepers, distributes load and helps restrain the track while allowing drainage and resilience. Other forms use a concrete slab or embedded support rather than a conventional ballast bed.

## 5.2 Rails and fastenings

Two rail forms are **bullhead** and **flat-bottom** rail. Bullhead rail has a relatively narrow base and is held in chairs, traditionally by keys. Flat-bottom rail has a broad foot and uses suitable baseplates or direct fastenings, including spring clips.

Rail section designations can include a mass per unit length. Keep the units attached to the number: a designation expressed in kilograms per metre cannot be compared numerically with one expressed in pounds per yard without conversion. UIC 54 and BS110A illustrate the distinction between metric and imperial rail designations.

Sleepers may be timber, concrete or steel. Their purpose is more than carrying the rail vertically: they maintain the relationship between the rails and transfer lateral and longitudinal forces into the supporting structure. Electrical insulation within the fastening arrangement can be essential where the rails form a track circuit.

## 5.3 Ballasted and slab track

Ballast must support the sleepers firmly without making the track a completely rigid structure. Tamping consolidates the ballast under and around sleepers. Trackside cables, rail connections and point equipment occupy some of the same working area, so track maintenance has important S&T interfaces.

Slab and embedded track use different support arrangements. One tunnel arrangement places concrete support blocks in resilient boots, allowing the support to accommodate traffic loads without behaving as an unyielding mass. The underlying lesson is that a track assembly needs controlled geometry and suitable resilience, whatever its form.

**i** In practice

When looking at a signalling item fixed to the railway, identify what holds it in its intended position. A detector attached to a rail, a machine mounted beside points and a cable fixed to a sleeper have different mechanical reference points and different exposure to track work.

## 5.4 Rail joints and continuous welded rail

Fishplates join rail ends mechanically. A conventional joint creates a discontinuity that must be maintained. Welding reduces the number of joints and changes how thermal expansion is managed within the track structure.

Temperature changes still matter in welded rail. Expansion and contraction are restrained, so longitudinal forces develop. Track design, stressing and maintenance govern those forces. The applicable stressing requirements must be established for the site.

For signalling, the electrical function of a joint is a separate question from its mechanical function. An ordinary joint may need a bond to provide a dependable electrical path. An **insulated rail joint** intentionally interrupts that path. A glued insulated joint combines mechanical strength with electrical separation and can be incorporated into welded track.

The question to ask is therefore not simply “is there a joint?” but “what continuity or insulation is intended here?” The train-detection chapter develops the consequences.

## 5.5 Gauge and the wheel-rail interface

**Track gauge** is the distance between the specified inner faces of the rails. Standard gauge is 1,435 mm. It is distinct from **loading gauge**, which concerns the space available for a vehicle and its load.

A wheelset has two wheels connected by an axle. The wheel treads are profiled rather than being simple cylindrical surfaces. A lateral displacement changes the effective rolling radii at the contacts, contributing to guidance. Flanges provide an important boundary to movement, but ordinary guidance cannot be explained merely as flanges continuously rubbing against the rails.

At points and crossings the wheel-rail relationship becomes more complex because routes diverge and flangeways interrupt parts of the running surface. This is why the next chapter introduces check rails and crossings as well as moving point blades.

## 5.6 Relating the track to its plan

Railway basics explains how tracks are represented on plans, how lines are named and how the spaces around them are described. Use those references to identify the track before considering its construction or the equipment attached to it.

## 5.7 Activity: a joint has two jobs

A rail joint is mechanically sound, but its insulating components no longer provide the intended electrical separation. Explain why a satisfactory mechanical inspection alone cannot establish that the signalling arrangement is correct.

> **i** Suggested reasoning
>
> The joint supports and connects the rail ends mechanically, but its signalling purpose may be to separate adjacent track circuits. Losing insulation can connect circuits that should be electrically separated. Mechanical condition and electrical function must both be understood.

## 5.8 Key points

- Track supports and guides the vehicle and can form part of signalling circuits.
- Mechanical continuity and electrical continuity are different requirements.
- Track geometry and support affect equipment attached to the railway.
- Track gauge and wheel profiles govern the wheel–rail relationship.

# 6. Points

**i** Learning objectives

Identify the main parts and rail forms of points; distinguish normal/reverse from facing/trailing; explain electric, EP and hydraulic operation; compare combined machines; and relate locking, detection, back drives and insulation to the complete layout.

## 6.1 Selecting a physical path

Points allow one track to divide into two, or two tracks to converge into one. The fixed outside rails are **stock rails**. The moving rails are **switch blades**, also called point blades. The moving end is the **toe**. Moving the blades closes one against its stock rail while opening the other to provide a flangeway.

Stretcher arrangements maintain the required relationship between the blades. Beyond them, the **crossing** provides the intersection of the rail paths. A fixed crossing includes gaps needed for wheel flanges. **Check rails** guide the wheelset through this area by acting against the back of a wheel flange, helping the opposite wheel follow the intended path at the crossing nose.

A turnout is therefore a coordinated assembly of moving and fixed track components. The machine is only one part of it.

## 6.2 Rail form and switch fit

The stock rail can be bullhead or flat-bottom and can have an inclined or vertical arrangement. Switch-rail profiles include undercut, straight-cut and chamfered forms. The mounting and fittings must suit the actual rail and switch combination.

Where the stock rail is inclined, the switch rail need not have the same inclination. Looking only at the visible head or assuming that every turnout has the same section can therefore give an incorrect picture of blade fit and detector geometry.

Closure concerns the relationship of the closed switch to its stock rail. Opening concerns the space provided between the opposite switch and stock rail for the wheel flange. Both matter. A blade that is held clear at its toe does not automatically establish the correct shape and clearance over its whole working length.

## 6.3 Two independent descriptions

**Normal** and **reverse** identify the two designated positions of the points. The designation is recorded on the plan. Normal must not be inferred merely because a route looks straight or appears to carry more traffic.

**Facing** and **trailing** describe the direction of a movement relative to the turnout. A facing movement approaches the toes and encounters the divergence. A trailing movement approaches from one of the converging tracks.

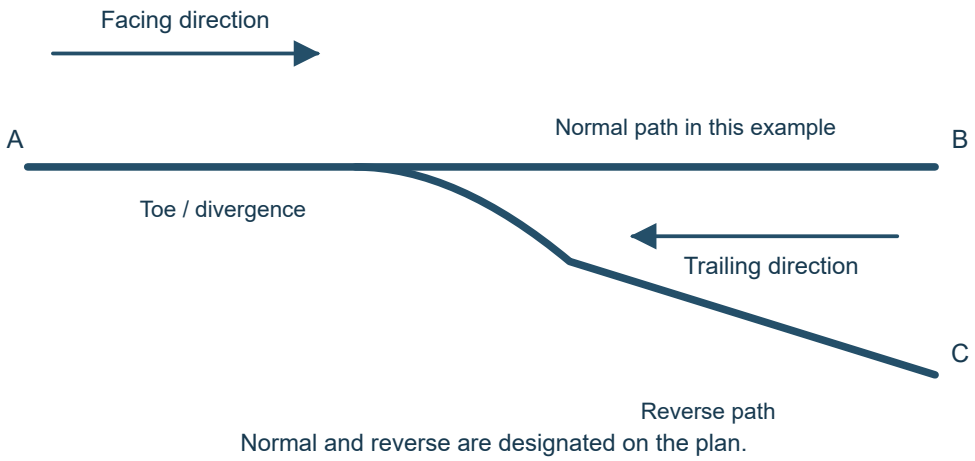



Figure 6.1: Normal/reverse describe position; facing/trailing describe movement direction. This example designates A-B normal.

| Movement in the illustrated example | Direction description | Required position |
|---|---|---|
| A to B | Facing | Normal |
| A to C | Facing | Reverse |
| B to A | Trailing | Normal |
| C to A | Trailing | Reverse |

Changing the normal designation on the plan would change the last column, not the facing/trailing descriptions.

## 6.4 Crossovers, diamonds and slips

A **crossover** combines two point ends to connect adjacent tracks. The two ends may be operated under one identity. Their detection must establish the conditions required for the combined movement; seeing one end correctly positioned does not establish the state of the other.

A **diamond crossing** lets two tracks cross at the same level without providing a route from one into the other. A **switch diamond** replaces parts of the fixed crossing arrangement with moving rails. A **single slip** adds a connection around one side of a diamond; a **double slip** provides connections around both sides. A **scissors crossover** combines crossovers within a compact length and includes a crossing between them.

A **swing-nose crossing** uses a movable crossing nose to close the required running path. Where moving crossings are provided, the blades and crossing equipment must be correctly coordinated and proved for the route.

The learning task is to trace each possible path before counting point machines. Complex trackwork may have several moving parts serving one movement.

## 6.5 Catch points and trap points

Spring catch points are intended to derail vehicles running back down a gradient. Trap points protect a running line against an uncontrolled movement from a siding or other connected line.

A trap arrangement provides protection by directing an unauthorised movement away from the line being protected. The derailing position can be its designated normal position. The fact that it leads away from usable track does not make it a mistaken setting.

These arrangements show that the safety purpose of a set of points may include protection from a movement as well as routing an authorised one.

## 6.6 Drive, lock and detect

Power-operated points have three functions that must be distinguished:

| Function | Purpose |
|---|---|
| Drive | Supply and transmit the force to move the blades |
| Lock | Restrain the required parts in their established position |
| Detect | Provide the electrical evidence required by the interlocking |

In a typical electric point machine, a motor drives gearing and a mechanism that converts rotary motion into linear movement. A clutch limits the consequences of an obstruction. A lock mechanism follows a sequence of unlocking, moving and relocking. Separate detector connections follow the blades and operate the detection contacts.

The machine arrangement described here has four mechanical connections: one drive, one lock and a detector connection for each blade. They do different jobs. Motor rotation or completion of part of the drive stroke is not enough to prove that the whole turnout has reached the required state.

The mechanical reference is also important. If detection depends on the relationship between blades and a machine, the mounting and connection between the machine, rails and track support are part of the complete arrangement.

> ! Detection is evidence with a defined meaning
>
> A detection circuit is intended to prove specified mechanical conditions. Its validity depends on the complete arrangement, including its adjustment, connections and reference points. An indication is not a substitute for understanding those conditions.

## 6.7 Methods of operation

### 6.7.1 Combined and non-combined electric machines

A **combined machine** brings the drive, locking and detection mechanisms into the machine assembly. External connections associate those mechanisms with the switch rails and stretcher arrangements.

A **non-combined machine** provides the movement while the lock and detector are separate items. In an SGE HA arrangement, separate operating connections drive the switches and a remote facing point lock, with a separate detector following the required conditions.

The distinction is about where the functions are implemented. Non-combined does not mean that the turnout can dispense with locking or detection. It means the complete equipment layout must be followed beyond the motor housing.

## 6.8 HW machines

An HW mechanism uses a motor, gearing and a bevel gear to operate a throw bar through an escapement. A roller moving in the shaped escapement slot produces phases in which the lock mechanism moves while the throw bar remains stationary, followed by the movement of the switches.

The sequence has three functional stages:

1. **Unlock:** the mechanism withdraws the lock dog from the established lock notch.
2. **Drive:** the throw bar transfers the switch rails to the opposite lie.
3. **Relock and prove:** the lock dog enters the other notch and the detector mechanism establishes the specified final conditions.

Lock blades follow the lock connection, while detector blades follow the switches through their individual rods. Their final relationship is what allows the detection contacts to provide the required detection indication. The motor reaching a particular rotation cannot substitute for that relationship.

The HW 1000 and HW 2000 arrangements have different motor and clutch designs. The HW 1000 uses a split-field motor with a friction clutch in the arrangement described here; the HW 2000 uses a permanent-

magnet motor and electromagnetic clutch. The clutch limits transmitted torque when movement is obstructed, but correct functioning still depends on the specified control and protection.

## 6.9 Electro-pneumatic points

An electro-pneumatic (EP) machine obtains its operating force from compressed air. Electrical valves control that force. Loss of electrical control and loss of air pressure can therefore produce similar final symptoms through different paths.

A typical SGE arrangement includes an air motor, valve unit, escapement mechanism and detector. The air motor is a piston and cylinder. Applying pressure to one side while exhausting the other produces the required direction of movement.

The electrical needle valves provide the normal, reverse and lock-control functions. They operate control-air paths which in turn operate the main air valves. This lets relatively small electrical devices control the larger air flow needed by the motor.

In the normal sequence, the direction is selected and the lock-control function enables the air supply. The escapement coordinates unlocking, switch movement and relocking. When the required final conditions are detected, the operating pressure is removed and the cylinder is exhausted as designed. Mechanical locking preserves the established position; continuous air force is not a substitute for the lock.

A separate point detector follows both switch rails and the lock slide in the arrangement described. The supply reservoir, pipes, valves, piston, mechanism and detection connections are all relevant to an account of the movement.

### 6.9.1 Worked example: valves called, no final detection

An EP direction command and lock command are registered, but the points do not reach valid detection. Several explanations remain possible: insufficient air supply, failure to admit or exhaust air, resistance in the mechanism, incomplete blade movement or a missing detection indication. The commands alone cannot distinguish them.

The useful study exercise is to follow **electrical command → valve action → air flow → piston movement → escapement and switch movement → locking and detection**. A missing final indication does not identify the detector as the only possible source of the failure.

## 6.10 Clamp locks

A clamp lock arrangement uses hydraulic power to move drive-lock slides. The sequence first unlocks the closed blade, then transfers the blades, then clamps the newly closed blade to its stock rail. Detection is associated with the rail-mounted mechanism and includes conditions for both the closed and open blade.

This gives another implementation of the same functional sequence. It also shows why mechanical support, rail attachment and electrical insulation are important interfaces between permanent way and S&T.

The introductory comparison concerns operating principles. Product-specific adjustment values, motor run times, manual operation and tests belong in the relevant controlled equipment documentation.

### 6.10.1 Components of the clamp-lock arrangement

A rail clamp-lock installation has a lock-and-detector mechanism associated with each side, drive-lock slides, actuators, a coupling arrangement and an electro-hydraulic power pack. The lock body references the stock rail, while the switch carries the parts that move with it.

The power pack drives the selected hydraulic actuator while allowing fluid from the opposite side to return. Initial slide movement releases the closed switch's lock arm. Continued movement transfers the switches. Final movement raises and supports the newly closed side's lock arm so that the blade is clamped and the required detection can be established.

The stretcher and coupling arrangements have distinct roles. In the mechanism described, the stretcher transfers the other switch while the coupling coordinates the drive-lock slides and the retracting actuator. Calling every connecting bar a drive rod obscures that sequence.

Track movement can change the relationships on which the mechanism depends. **Road spread** concerns a change in the relationship between the stock rails; **switch creep** is longitudinal movement of the switch relative to its reference position. Rail attachments and the permanent-way condition therefore contribute to the equipment's performance.

## 6.11 Back drives and supplementary detection

A long switch requires controlled movement beyond the toe. A **back drive** transmits or supplies movement farther along the switch so that the closed rail fits and the open rail provides the required clearance over its working length.

Mechanical back drives use rods, cranks and supported connections. Hydraulic arrangements use additional actuators associated with the power pack. The number and positions depend on the turnout design; they cannot be inferred from the toe machine alone.

Supplementary detectors can provide detection indications farther along the switch. Toe detection and a back drive perform different functions: one reports defined conditions at its connection, while the other controls movement elsewhere. A satisfactory toe indication cannot automatically establish every condition along a long switch.

## 6.12 Insulating the point fittings

A metal stretcher or sole plate can create an unwanted electrical connection between rail paths. Insulation within the fittings preserves the separation needed by the track circuit while retaining the mechanical connection.

Insulated arrangements can include divided sole plates and stretchers, bushes, ferrules, washers and insulation pieces at rods or fittings. An insulated ferrule separates the bolt from the surrounding metal; an insulating washer separates faces. Replacing only one part of that separation can leave another conductive path.

The required points include the switch and lock connections, detector rods and drive connections appropriate to the layout. A mechanically secure repair can still be electrically wrong if its insulation is defeated. Relate the fittings to the bonding arrangement as well as to the point machine.

## 6.13 What lock and detection tests establish

A **facing point lock test** establishes the specified relationship between switch fit and lock engagement. A **detection test** establishes the required relationship between the actual switch and lock conditions and the electrical indication. They address different links in the chain.

Testing has both safety and reliability purposes. The detection must not indicate the required position outside its permitted mechanical conditions, while the equipment must operate reliably within its permitted working tolerance. Setting everything as tight as possible can produce unreliable operation or mask the intended sequence.

The test positions, gauges, preparation and acceptance values depend on the mechanism and applicable specification. A point machine and a rail clamp lock do not necessarily use the same means of separating lock indication from switch detection during a test. Those procedures must be learnt from the equipment-specific instructions and carried out under the required authority.

## 6.14 Worked example: one end detected

A crossover is commanded reverse. End A is reported reverse, but end B has no valid detection. Can the signal clear simply because the first end is correct?

No. The route depends on a continuous, correctly established path through both ends. If the required detection includes both ends, the interlocking is missing part of its evidence. A successful drive command and one end's indication cannot stand in for the other end's indication.

## 6.15 Key points

- Normal/reverse are designated positions; facing/trailing describe movements.
- Points and crossings include fixed rails, moving rails and guidance through crossings.
- Drive, locking and detection are separate but coordinated functions.
- Combined, non-combined, EP and hydraulic equipment implement those functions in different arrangements.
- Back drives and supplementary detection address conditions beyond the switch toe.
- Trap points protect a line from an unwanted movement.
- The mechanical reference and electrical insulation form part of the signalling arrangement.

# 7. Lineside signals

**i** Learning objectives

Explain colour-light signal operation; relate aspect sequences to braking and headway; distinguish aspect selection, filament monitoring and lamp proving; describe route and subsidiary indications; and explain the operating principles of AWS and TPWS.

## 7.1 Communicating with the driver

An **aspect** is the indication displayed by a signal. Colour-light signals distinguish aspects by colours or combinations of colours. Position-light signals also use the arrangement of the lights. The meaning must be taken from the applicable signalling system, rather than from colour alone.

In route signalling, signals communicate movement information and route indications identify a divergence or destination. The driver also needs knowledge of the route and its speed restrictions. This differs from systems that use aspects to communicate specified speed categories directly.

A signal at its most restrictive aspect is said to be **on**. When it displays a less restrictive aspect it is **off**, or has **cleared**. Clearing does not necessarily mean displaying green: a signal displaying yellow has also cleared from red.

## 7.2 Warning before a stop

A sequence of red/green stop signals needs an associated means of warning the driver about a red signal early enough to stop. Yellow/green distant or repeater signals can provide this warning in association with the stop signals.

The governing relationship is between the point at which the driver receives the warning and the location where the train must stop. Braking requirements depend on the trains and the railway, including gradient. Visibility of the red signal alone may not provide sufficient stopping distance.

A three-aspect system combines stop and warning functions at successive signals:

| Aspect in the simplified sequence | Information conveyed |
|---|---|
| Red | Stop |
| Single yellow | Be prepared to stop at the next signal |
| Green | The next signal is displaying a proceed aspect within this sequence |

A four-aspect system adds double yellow as an earlier caution in the sequence. A simplified progression towards a signal at red is green, double yellow, single yellow, red.

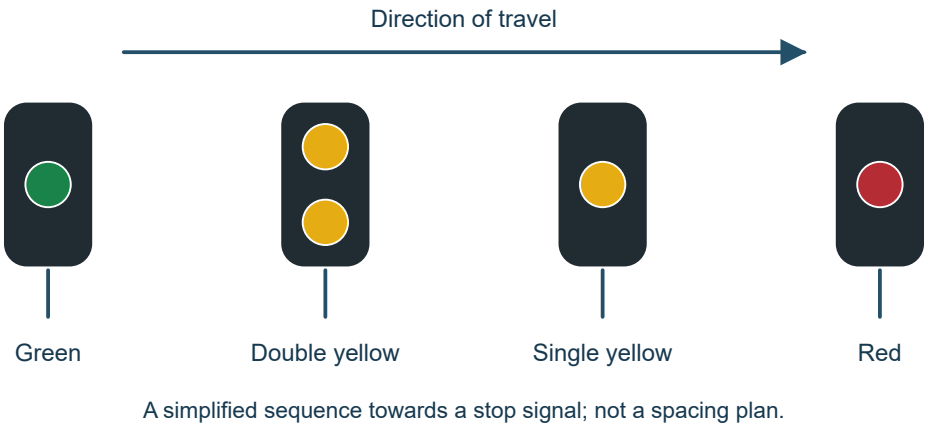



Figure 7.1: A simplified four-aspect warning sequence, read in the direction of travel. Actual spacing and controls depend on the design.

This is a principle diagram. It does not cover flashing aspects, special sequences or every possible installation.

## 7.3 Overlaps

An **overlap** provides a defined distance beyond a stop signal as part of the protection for a train approaching that signal. In a simple plain-line arrangement, the signal in rear requires the appropriate section and overlap beyond the next stop signal to be clear.

The overlap is additional protection for an overrun; it is not an extension of the driver's authority to pass the stop signal. It can include points and more than one train-detection section. Those conditions must be dealt with by the interlocking.

## 7.4 Headway

**Headway** is the separation between successive trains, expressed as a distance or as a time at a stated speed. A minimum-distance headway calculation must include the stated assumptions, such as train length, overlap, signal sighting allowance and braking distance.

For a simplified, evenly spaced plain-line arrangement:

$$H_3 = L + O + 2B + S \tag{7.1}$$

$$H_4 = L + O + 1.5B + S \tag{7.2}$$

Here $H_3$ and $H_4$ are the three- and four-aspect headway distances, $L$ is train length, $O$ is overlap length, $B$ is braking distance and $S$ is a sighting allowance, all expressed in the same distance unit. These simplified teaching relationships depend on the stated assumptions; they are not general design rules for all layouts.

For an invented example, take $L = 200$ m, $O = 180$ m, $B = 1,000$ m and $S = 200$ m. The results are $H_3 = 2,580$ m and $H_4 = 2,080$ m. At a constant 20 m/s, these distances correspond to 129 s and 104 s respectively. The reduction is 500 m, or 25 s at that speed. Actual service planning also depends on junctions, station stops, varying speeds and other constraints.

Adding intermediate signals does not halve every component of headway. Train length, overlap and sighting allowance remain in this simplified comparison.

## 7.5 Inside a filament colour-light signal

In a separate-aperture filament signal, each aspect has a compartment containing its lamp, holder, electrical connections and optical system. Internal partitions prevent an illuminated lamp from lighting the lens of another aspect. The rear access and sealing arrangements protect these relationships from dirt, water and disturbance.

A double-filament lamp has a **main** and **auxiliary** filament with the required contacts and common connection. The holder fixes their position relative to the optical system. The correct lamp type and orientation matter: two lamps with similar bases can have different filament geometry or ratings.

The optical assembly collects and directs the light into the required beam. A long-range lens concentrates the light; a spread arrangement provides a wider viewing field where required. A **hot strip** directs part of the light towards the position of a driver close to the signal. Beam alignment is therefore more than pointing the outside of the housing towards the track.

Hoods, backgrounds and control of reflections support recognition of the intended aspect. A **phantom aspect** is an apparent illumination caused by extraneous light, such as sunlight, rather than the intended lamp. Maintaining sighting and optical integrity preserves the meaning of the electrical control.

### 7.5.1 The electrical supply to the lamp

A filament arrangement can send an a.c. supply along the tail cable and step it down at an aspect transformer in the signal head. This reduces the current required in the longer cable compared with distributing the same power at the lower lamp voltage. The transformer and lamp connections still have to provide the specified conditions at the filament.

Separate tappings or connections can set the intended main and auxiliary operation. The lamp's nominal marking is not necessarily the specified operating value in the complete assembly. Adjustments, beam setting and holder position are governed by the applicable equipment instructions.

## 7.6 Aspect selection, filament changeover and lamp proving

Three functions must be distinguished:

| Function | Question answered |
|---|---|
| Aspect control | Which aspect has been selected by the signalling logic? |
| Filament changeover and monitoring | Is the main filament available, and has the auxiliary been brought into use? |
| Lamp proving | Is the required aspect-current evidence present for the signalling controls? |

In a simplified four-aspect relay arrangement, the following states produce the normal sequence:

| Aspect | HR | HHR | DR |
|---|---|---|---|
| Red | Down | Down | Down |
| Single yellow | Up | Down | Down |
| Double yellow | Up | Up | Down |
| Green | Up | Up | Up |

The control logic enforces the intended progression. It does not normally allow an arbitrary combination such as DR up with HR down. These names and states explain the particular arrangement; other signal technologies implement the functions differently.

A local lamp relay can be energised by the main-filament current. If that filament opens, the relay releases and switches the supply to the auxiliary filament. Monitoring can report that the main filament has failed even though the signal remains illuminated.

The **ECR** lamp-proving function provides separate evidence used by the signalling controls. Its slow-release characteristic can bridge the short interruption during a legitimate aspect change or filament changeover, while a sustained loss removes the corresponding lamp-proving indication. The specified delay and proving path are part of the circuit design.

In the four-aspect arrangement described, the second yellow has a different proving treatment from the red, first yellow and green. Losing only the second yellow changes double yellow to the more restrictive single yellow. That example should not be extended to every multi-lamp display: some missing lamps can make a route indication ambiguous rather than simply more restrictive.

### 7.6.1 Worked example: a lamp is lit but a filament alarm remains

A signal continues to display its selected aspect after its main filament fails. The auxiliary is carrying the required current, but a filament alarm is present. These observations can be consistent: changeover has preserved the display while monitoring reports the loss of the main filament.

The alarm and lamp-proving functions answer different questions. A visible light does not prove that both filaments are intact, while a main-filament alarm does not necessarily mean that the signal is dark. Read the particular lamp and proving circuit before interpreting either indication.

## 7.7 Searchlight signals

A searchlight signal uses a lamp and an optical path with a moving coloured **roundel**. Its mechanism selects red, yellow or green within that path. In the arrangement considered here, the released mechanism selects red; energisation with one polarity selects yellow and the opposite polarity selects green.

Mechanism contacts repeat the roundel's position. Lamp-current proving and roundel-position proving are different evidence: a lamp can be illuminated while the colour-selection mechanism fails to reach its commanded position. A four-aspect searchlight arrangement adds a separate upper yellow to provide double yellow.

The lamp, reflector, roundel and lenses form the optical system. A reflector designed into that system must be understood with its associated controls and optical design, rather than applying a rule inferred from a different signal head. Main and auxiliary filament arrangements also have to maintain the intended optical alignment.

## 7.8 Route indicators and signal sighting

A junction direction indicator, often called a **feather** or **route arm**, identifies a diverging route using an illuminated row of white lights. A theatre-type indicator displays a character identifying a route or destination. Their uses depend on visibility and operating context.

A route indication complements the signal aspect. It does not replace the need for the route to be correctly established. The controls must associate the indication with the route actually proved.

Physical signal design must make the intended display recognisable. Hoods, backgrounds, lens alignment and suppression of misleading reflections contribute to this. Plan symbols represent the signal's function and orientation; the exact physical installation requires more information than the scheme-plan symbol alone.

### 7.8.1 Proving the route display

A junction indicator must produce a recognisable indication before the associated main aspect is permitted where route-display proving is required. A relay proving circuit can assess the selected lamp group through its designed current conditions. The number of lit lamps and their contribution to recognisability are equipment-specific.

A theatre-type display forms a character by illuminating selected lamp groups or an optical pattern. Some lamps are shared between different characters. Its controls must select the correct combination and obtain the required lamp-proving indication; finding current in any one lamp does not prove the intended character.

A stencil displays a fixed legend when illuminated. Alternative stencils need controls that prevent conflicting legends appearing together. Fibre-optic arrangements can carry light from a lamp to the positions that form the displayed pattern. The control path, light source, optical path and visible legend are distinct links in that chain.

## 7.9 Shunt and subsidiary signals

Shunt signals authorise movements under a different set of conditions from a main running signal. A proceed shunt indication must not be assumed to prove the route clear to the same extent as a main running route. Some shunt movements are specifically intended to approach vehicles already standing on the line.

A **subsidiary signal** is associated with a main signal and can authorise a movement while the main aspect remains at danger. A **call-on** is a use that allows a train to enter an occupied line under the

applicable conditions, such as entering a platform to join another train. The interlocking conditions are specific to the movement.

Other arrangements include preceding shunt signals, yellow shunts, opposing-locking-omitted arrangements and limit-of-shunt signals. These names identify specific purposes; they are not interchangeable variants of a universal shunting permission. In particular, a limit of shunt defines the limit of the relevant movement rather than providing a proceed aspect.

## 7.10 Fitting a train into an occupied platform

**Lime Street control** uses the occupation of defined approach and platform sections to establish whether an approaching train can fit into the available space. Its logic depends on the relative lengths and occupation of those sections, not merely on a generic “platform occupied” indication.

Suppose the available platform section and a designated approach section have equal lengths. If the approaching train is standing at the signal but also occupies a further section behind that approach section, it is too long for the assumed space. The call-on conditions therefore exclude that state.

This is an example of translating a physical requirement into train-detection conditions. It must be understood from the actual layout and control requirements before its logic can be interpreted.

## 7.11 Automatic Warning System

AWS combines track equipment and trainborne equipment to provide an audible indication and a visual reminder. In a standard signal-associated arrangement, the train receiver encounters a permanent magnet with its south pole uppermost and then an electromagnet which presents the required north-pole field when energised.

The permanent magnet initiates the warning sequence. The subsequent energised electromagnet produces the clear indication under the required signal conditions. If that cancelling field is absent, the driver receives the warning and must acknowledge it; failure to acknowledge initiates a brake application.

The cab’s yellow-and-black **sunflower** or sunburst indication reminds the driver of an acknowledged warning. It is not a live repeat of the signal aspect: the signal can change after the train has passed the equipment. The driver still responds to the signals and route requirements.

For a permanent warning, such as a defined speed-restriction warning arrangement, a permanent magnet can be used without the signal-associated cancelling electromagnet. Bi-directional arrangements can require suppression and additional control of the magnetic fields so that the equipment has its intended effect for each direction.

### 7.11.1 The inductor circuit

A relay-controlled AWS electromagnet can include both the green aspect-control condition and lamp-proving condition in its feed. The circuit therefore needs more than an instruction to display green. Both feed and return paths and the specified proving conditions must be followed when interpreting it.

An inductor stores energy in its magnetic field. A suppression diode or rectifier across the coil limits the switching transient when the current is interrupted, reducing arcing at the controlling contacts in the arrangement concerned. Its polarity and circuit connections form part of the design.

The track mounting, protecting ramp, magnet identity, direction and relationship to the associated signal are also functional parts of the installation. Correct electrical operation alone does not establish correct positioning or mechanical clearance.

## 7.12 Train Protection and Warning System

**TPWS** provides defined brake-intervention functions through track loops and trainborne equipment. Its **overspeed sensor system (OSS)** uses an arming and trigger sequence on the approach. The elapsed time between the two is related to the train’s speed and the loop separation.

In a simple timing model, passing the trigger while the onboard timing window is still active initiates an intervention. Greater separation gives a longer transit time at the same speed, so loop position is a functional part of the designed threshold. A **train stop system (TSS)** uses a closely spaced loop arrangement at the protected location to initiate the defined stop intervention when active.

The signalling interface controls the required active state and returns the specified fault or health information. An instruction to activate is different from evidence that the installation is functioning. A fault indication must be interpreted through the actual interface and its designed signalling response.

TPWS addresses specified locations and conditions; it does not guarantee that every train will stop before every possible conflict. Performance depends on the layout, approach conditions, equipment and train braking. Loop spacings, timing values, suppression and tests belong to the applicable design and equipment requirements.

## 7.13 Key points

- Clearing means becoming less restrictive, not necessarily showing green.
- Warning distance must support stopping at the protected location.
- Four-aspect signalling adds an earlier warning stage; capacity depends on the whole arrangement.
- An overlap is protection, not movement authority.
- Shunt and call-on routes have purpose-specific controls.
- Aspect selection, filament monitoring, optical performance and lamp proving provide different evidence.
- AWS warnings and TPWS interventions depend on defined track-to-train arrangements.

# 8. Circuit reading

**i** Learning objectives

Distinguish signalling plans, circuit diagrams and bonding diagrams; interpret equipment names and supply labels; trace complete and branching paths; follow references between sheets and locations; and explain red and green alteration markings.

## 8.1 Drawings communicate different kinds of information

A signalling circuit is both a logical arrangement and a physical network of equipment, wires and terminals. Reading it requires you to understand what the circuit is intended to do and how the actual connections achieve that function.

Three drawing types answer different questions:

| Drawing | Main question | Typical contents |
|---|---|---|
| Signalling or scheme plan | What equipment is provided, and where does it apply? | Tracks, signals, point identities, train-detection sections and operating directions |
| Circuit or wiring diagram | How are the electrical functions connected? | Contacts, coils, supplies, terminals, cable cores and sheet references |
| Bonding diagram | How are the individual rails electrically connected or separated? | Insulated joints, rail bonds, jumper connections and traction-return arrangements |

A scheme plan generally uses one line for each track. A bonding drawing needs the individual rail paths. Neither can replace the circuit detail needed to identify a particular relay contact or cable termination.

A schematic explains a principle with reduced detail. A working wiring diagram also identifies the connections needed for the installation. Always establish what kind of drawing you are reading before deciding what information it proves.

### 8.1.1 Wiring layout and the written circuit

A wiring layout can group a relay's coil and contacts together, making the apparatus and its connections recognisable. A **written circuit** distributes the symbols into convenient feed-to-return paths, with function names identifying parts that belong to the same apparatus. Both can describe the same electrical connections while looking quite different on the page.

Converting between them is an exercise in preserving connections. Retain each terminal identity and junction, then follow the complete paths. Rearranging the symbols for clarity must not introduce a join where wires merely cross, omit a branch or exchange two contact terminals.

## 8.2 Start with the title block and index

The title block identifies the installation, drawing number, sheet and revision. An index helps locate related sheets. Read these before concentrating on the circuit: a familiar-looking drawing for a different installation or revision can give a convincing but incorrect account of the equipment in front of you.

Build a small map of the drawing set. Locate the equipment circuits, supply and looping sheets, cable information, and rack or location profiles. The diagram containing a relay coil is not necessarily the sheet containing every contact of that relay.

## 8.3 Symbol, identity and termination

A circuit element has three layers of information:

1. **Symbol:** what kind of electrical item or contact is represented?
2. **Identity:** which item performs this function?
3. **Termination:** exactly where does the connection enter or leave it?

For example, `AE TPR` identifies a track repeater relay associated with track circuit AE. `R1` and `R2` may identify its coil connections in a particular arrangement. A contact labelled `AE TPR A1-A2` belongs to that relay but represents a different pair of physical connections. The actual drawing determines the terminal designations; do not assume every relay uses the same coil pair.

The same equipment identity can occur many times because a relay has several contacts. Repeated identity does not mean that every symbol represents a separate relay.

### 8.3.1 Read the whole symbol

The contact mark alone may be insufficient. The associated armature symbol, position letters, terminal designations and any timing or polarity notation distinguish different kinds of contact operation. A special symbol or reference number can direct the reader to a drawing note that supplies an essential condition.

Read those notes before assigning a state. In particular, a polar contact marked normal or reverse must be interpreted through the relay's specified operation and retention behaviour. The relay contact and polar-relay explanations develop these distinctions. A neutral relay's simple up/down table cannot be applied to every symbol.

## 8.4 Reading function names

The final letter commonly identifies the apparatus, while preceding letters describe its function. The equipment number or track reference associates it with the railway item.

| Example | Interpretation |
|---|---|
| AE TR | Track relay for AE |
| AE TPR | Track repeater relay for AE |
| 23 ECR | Lamp-proving relay associated with signal 23 |
| 23 HR / HHR / DR | Yellow, double-yellow and green aspect-control functions |
| 345 NWKR / RWKR | Normal and reverse point-detection relays for points 345 |
| 345 NWKE / RWKE | Normal and reverse point-indication lamps for points 345 |
| WM | Point motor function in the applicable naming convention |
| T/J | Transformer-rectifier designation |

Letters have different meanings in different positions. An R can describe reverse or red within a name and identify a relay at its end. Parentheses or quotation marks can identify special functions. Use the drawing conventions and the relevant symbol and nomenclature reference rather than guessing an unfamiliar combination.

The relays chapter develops the relationship between coil state, front contacts and back contacts. A front contact is not automatically closed just because it is called normal in ordinary conversation.

## 8.5 Supply labels and protection

Supply labels follow their own convention. In a drawing using the following labels:

| Label | Meaning |
|---|---|
| B50 | Positive side of the identified nominal 50 V d.c. supply |
| N50 | Negative side of that supply |
| BX110 | Feed side of the identified nominal 110 V a.c. supply |
| NX110 | Return side of that a.c. supply |
| BN or BNX | An intermediate connection in the relevant supply arrangement |
| E | An earth connection where shown |

The number states a nominal voltage designation, not a measured result. B50 and N50 are the two sides of one labelled supply; the labels do not mean that each conductor has that voltage to earth. In an earth-free circuit neither side should simply be treated as an earth return.

Fuse identity and rating are separate information. `f007 3A`, for example, identifies fuse 007 with a 3 A rating. The rating is a protective-device characteristic, not the normal current drawn by every circuit it supplies.

## 8.6 Trace the whole path

Start at the identified feed and follow the path through protection, controls, the load and the return. At every contact, ask which relay or mechanism operates it and what state makes it conduct. At every continuation, follow the stated reference rather than assuming that the line ends there.

For a simplified repeater circuit, the path can be written as:

**B50 → fuse → AE TR front contact → AE TPR coil → second AE TR front contact → N50.**

Both contacts must be made for the repeater coil to receive current. The second contact controls the return side; it is not a second source of power. This is an example of double cutting. Removing either required contact path prevents energisation in the ideal circuit.

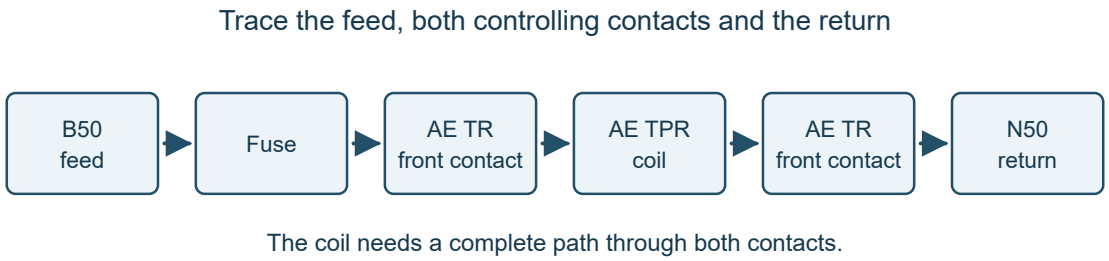



Figure 8.1: Reading a complete feed-to-return path. The boxes represent named circuit elements, not standard wiring symbols.

The drawn order across the page makes the path easy to follow, but a wire may physically pass through several terminal strips and a cable before reaching the next element. Logical proximity is not physical proximity.

## 8.7 Series, parallel and branching paths

Contacts in series all have to conduct for that path to be complete. Parallel branches provide alternative paths. A branch can also divide a shared feed between different loads, each with its own controls and return.

Do not confuse an alternative path around a contact with a separate load fed from the same junction. In the first case either branch may energise one coil. In the second, one branch can feed a lamp while another feeds a relay, and their remaining conditions may differ.

A **discrete circuit** has its own complete feed-to-return path. Several discrete circuits may appear on one sheet. Shared supplies can be shown through **positive looping** and **negative looping**, with a separate looping sheet identifying the fuse, connections and destination sheets. A circuit shown without the whole busbar arrangement still depends on those supply connections.

## 8.8 Follow the boundary between locations

Incoming and outgoing describe direction relative to a stated location. A control can leave the relay room for a point machine, while detection returns from the point equipment. A circuit that is outgoing at one housing is incoming at another.

Follow both sides of the circuit across the boundary. Record the location identity, terminal, cable identity, core and receiving terminal. A matching function name alone does not prove that you have found the correct conductor.

Interlinking cables connect equipment within a relay room or between racks and the operating equipment. An interface can therefore exist within a building as well as along the railway. A rack or location profile helps turn the electrical identity into a physical position.

## 8.9 Red and green alteration work

Alteration drawings distinguish the existing arrangement from work to be added or recovered:

- **Red** identifies new work or additions.
- **Green** identifies work to be removed or recovered.
- **Black** generally represents retained work, subject to the drawing's stated conventions.

An all-new sheet can be endorsed accordingly, while a sheet being recovered can carry an overall recovery marking. Complex work can use separate addition and recovery drawings. Colours must be read with the issue, stage, notes and endorsements; a greyscale print can conceal important distinctions.

Adding a connection to a spare contact of an existing relay does not necessarily mean installing a new relay. Likewise, removing wires from a contact does not necessarily mean recovering the relay. The markings identify what changes, down to the relevant connection or item.

> **i** Read an alteration as two arrangements
>
> First establish the retained circuit with the green work removed. Then trace the completed circuit with the red work added. This helps expose a bypass, a missing return or an unintended change of logic without treating a coloured line as an isolated instruction.

## 8.10 Worked example: following a missing indication

An indication lamp is controlled by a point-detection repeater in a relay room. Its drawing refers to a separate looping sheet for the feed and to an interlinking cable for the lamp on the operating floor.

A learner looks only at the repeater contact and concludes that a closed contact proves the lamp should be lit. The conclusion omits the fuse and shared feed, the interlinking cores, lamp connections and return. It also assumes that the correct physical contact has been identified.

A complete account names the supply, fuse, controlling contact, intervening terminals, cable cores, lamp and return. If any part is uncertain, the explanation is incomplete. This is a drawing-reading exercise, not authority to disturb or test the operational circuit.

## 8.11 Key points

- Different drawings represent the railway, the logic and the individual rail or wire connections.
- Identify the symbol, equipment function and terminals separately.
- Trace the complete path, including shared supply sheets and returns.
- Incoming and outgoing have meaning only relative to a location.
- Red and green describe alterations; drawing status and stage determine how to interpret them.

# 9. Relays and relay logic

**i** Learning objectives

Explain relay construction and front/back contacts; distinguish neutral, biased, polar, latched, twin, timed and vane arrangements; interpret independent and dependent contacts; locate a relay through its records; translate logic into contact paths; and distinguish electrical holding from magnetic retention.

## 9.1 A switch operated by an electromagnet

A relay combines a coil, a magnetic circuit, a moving armature and electrical contacts. Coil current produces a magnetic force that moves the armature. The movement operates several contacts, allowing one circuit to influence other circuits without directly joining all their electrical supplies.

A relay is **energised** when its coil is supplied as required, and **de-energised** when it is not. In the ordinary neutral-relay example, it **picks up** when energised and **drops away** when the supply is removed. Special relays can have different behaviour, so the type matters.

The contact state is defined relative to the relay's operated state:

| Relay state in the basic example | Front contact | Back contact |
|---|---|---|
| Down | Open | Closed |
| Up | Closed | Open |

The words front and back avoid the ambiguity of saying "normally open" when a relay may normally be kept energised in service. A track relay, for example, can be energised throughout the time its section is clear.

## 9.2 Independent and dependent contacts

An **independent contact** has its own arm connection and associated fixed contact. A **dependent contact arrangement** uses a common arm that transfers between two fixed contacts. In the basic neutral example, these are the front and back contacts. The common terminal is shared, so the two paths must not be read as isolated switches.

Terminal notation depends on the relay construction. Some arrangements identify the finger and its arm, front and back connections; miniature plugboards use their specified letter-and-number positions. Translate the symbol through the applicable terminal information rather than transferring a familiar numbering rule to a different relay.

Contact behaviour during movement also matters. A specified **bridging contact** makes both paths during part of its travel. This is a particular characteristic, not something to assume for every changeover contact. Similarly, two breaks within one contact arrangement describe its physical interruption; they do not by themselves establish the separate feed-and-return control of a double-cut circuit.

## 9.3 Inside a miniature signalling relay

The coil is wound around a soft-iron core. The core, yoke and armature provide a magnetic path. Soft iron carries the flux readily but retains relatively little magnetism after the coil current is removed. The armature pivots towards the pole face and operates a drive bar, transferring movement to the contact arms.

| Part | Contribution to operation |
|---|---|
| Coil and core | Convert coil current into magnetic flux |

| Part | Contribution to operation |
|---|---|
| Yoke | Complete much of the magnetic path |
| Armature and pivot | Turn magnetic attraction into mechanical movement |
| Drive bar | Operate the contact arms |
| Buffer bar | Limit the contact movement and support the intended contact pressure |
| Residual pin | Maintain a small non-magnetic gap at the attracted position |
| Case and baseplate | Support and protect the assembly |

The **residual pin** prevents full metal-to-metal closure of the magnetic path. The resulting gap helps ensure that residual magnetism does not hold the armature attracted when the supply is removed. It has a different purpose from the contact buffer and from any return spring.

Contact materials are selected for the required resistance, current duty and resistance to welding. A miniature signalling relay can use silver and silver-impregnated graphite at its contact surfaces. Mechanical clearance, pressure and material choice are part of the complete relay design; a contact is more than an ideal switch drawn on paper.

For a neutral d.c. relay, reversing the coil current reverses the magnetic poles but still attracts the armature. A biased relay deliberately adds a directional characteristic. This distinction explains why observing an apparently correct voltage magnitude is insufficient without understanding the relay type and polarity.

## 9.4 Plugboards, wiring and equipment identity

The relay's spring connections engage the plugboard connections. Wires terminate in spade connectors retained in the plugboard, with identification beads linking the physical wires to their circuit records. Retaining arrangements secure the relay in position, while seals protect the internal assembly against unauthorised disturbance.

A connection can be electrically specified on a circuit sheet but physically located through a rack or location profile. A typical profile records the function, position, relay specification, coding and contact arrangement. The profile's stated viewing direction matters: a front-view arrangement cannot be transferred directly to a rear view without accounting for the change of perspective.

The number of wires accommodated by a connector is limited. Where several points must be joined, a designed chain of connections can distribute the common path instead of crowding all wires into one terminal. A drawn branching path therefore has a physical termination arrangement behind it.

A replacement relay must match the specified function and characteristics, including its voltage, timing, contact arrangement and coding. Pin coding is a safeguard against incorrect substitution; it is not a substitute for checking the label, records and physical condition.

## 9.5 Types and construction

The BR miniature relay family includes several special functions:

- A **neutral relay** provides the basic electromagnetic switching function.
- A **biased relay** is designed to respond to the intended current direction.
- An **a.c.-immune relay** is designed to operate on its specified d.c. supply without responding to the specified a.c. interference.
- A **contactor** has contacts suited to heavier loads, such as point-machine control.
- A **slow-pick relay** delays the operation of its contacts after energisation.
- A **slow-release relay** delays contact release after removal of coil current.
- A **latched relay** retains its operated state until the required release action.

## 9.6 Special relay behaviour

### 9.6.1 Double-wound and twin relays

A **double-wound relay** has two electrically separate windings acting on one mechanism. Either winding can operate the common contact assembly in the intended arrangement. The coil circuits are separate, but the resulting mechanical state is shared.

A **twin relay** contains two independent relay mechanisms in one housing, each serving its own function and contacts. It saves space where each function needs fewer contacts. Two coils visible in a case therefore do not, by themselves, tell you whether the device is double-wound or twin.

### 9.6.2 Immunity and timing

A copper sleeve or slug on the magnetic circuit opposes rapid changes of flux. Its position and the associated magnetic arrangement can create the required immunity or delay characteristics. An a.c.-immune d.c. relay is designed not to respond to the specified alternating interference; the characteristic has to match the electrical environment.

Slow pick and slow release delay different transitions. A track repeater can require a sustained clear input before picking. A lamp-proving relay can bridge the brief interruption of current during an intended change of aspect. Neither delay should be assumed from the relay's external shape or from a generic delay quoted for another type.

Dedicated timing functions can use different mechanisms:

| Timer | Principle |
|---|---|
| Synchronous motor | A motor and gearing move through a defined travel before operating contacts |
| Thermal | Heating and cooling of an element provide a controlled contact sequence |
| Electronic | A defined electronic timing process controls the output function |

The timing interval may include more than the first visible movement. In a thermal arrangement, the warm-up phase can start a sequence whose final output follows the cooling phase. Read the complete circuit before deciding when the timed function should occur.

### 9.6.3 Magnetic memory and electrical memory

A **magnetically latched relay** can retain its contact state without continuous coil current. A separate operation releases or changes it. The polar-stick example in the next subsection uses a particular form of this retention.

This differs from the electrical stick circuit described later in this chapter. An ordinary relay in an electrical stick circuit needs continuing energy through its holding path. Magnetic retention can preserve a state through a loss of supply, so the expression "no coil current means all relays are down" is unsafe as a general rule.

### 9.6.4 Two-position and three-position polar relays

A polar relay uses current direction to select normal or reverse. The table compares a retaining two-position arrangement with a three-position arrangement that has a centre state. The polarity-to-terminal relationship comes from the relay and circuit information.

| Input condition in the teaching comparison | Retaining two-position polar relay | Three-position polar relay |
|---|---|---|
| Specified normal polarity | Normal | Normal |
| Specified reverse polarity | Reverse | Reverse |
| No coil input | Last selected position retained | Centre; neither outer contact made |

Thus an unpowered polar relay cannot be interpreted from the word *polar* alone. The number of positions and the specified retention characteristic determine what its contacts can do.

### 9.6.5 Contactors and vane relays

A contactor uses contacts and interruption arrangements suited to a heavier load. Magnetic blow-out helps control an arc when a loaded contact opens. A contact suitable for an indication circuit cannot automatically interrupt a point-motor load merely because both are drawn as contacts.

An **a.c. vane relay** works differently from the neutral d.c. relay. Local and control magnetic systems act on a conducting vane. Their currents must have the intended frequency and phase relationship to produce the operating torque. In the double-element arrangement, loss of either required input removes that torque.

This gives a useful link to the train-detection chapter: the receiver must recognise the intended electrical evidence, rather than simply respond to any current present at the rails.

## 9.7 Worked example: two coils, two possible meanings

A drawing shows two coil pairs associated with a single housing. One learner assumes that the two functions have independent contacts. Explain what must be checked.

A double-wound relay has two windings operating the same mechanism; a twin relay has two mechanisms. The specification, naming, connection information and profile establish which arrangement is present. Counting coils or cases is insufficient to identify the logical independence of the functions.

## 9.8 From a contact to a complete path

Use the circuit-reading method to establish supplies, terminals and sheet references before translating contacts into logic. The examples below assume that the required feed and return are available, so they can concentrate on the relationships between relay states.

## 9.9 AND and OR

For an ideal teaching circuit, assume a suitable supply and no faults. To energise relay R when A **and** B are up, place front contacts of A and B in series. Both contacts must close to complete the path.

To energise R when A **or** B is up, place the front contacts in parallel. Either path can complete the circuit. OR is inclusive: both may be up.

| A up | B up | R for A AND B | R for A OR B |
|---|---|---|---|
| No | No | Down | Down |
| No | Yes | Down | Up |
| Yes | No | Down | Up |
| Yes | Yes | Up | Up |

A back contact introduces a condition involving the relay being down. For example, an A front contact in series with a B back contact expresses **A up AND B down**.

Be precise about “only when” and “whenever”. Saying that R may be up only when A is up makes A a necessary condition. It does not, by itself, say that A being up is sufficient: the circuit might contain other conditions. When drawing a complete truth table, state that the listed inputs fully define the teaching circuit.

## 9.10 Reading a compound requirement

Suppose the full energisation requirement is:

**R up = A up AND (B up OR C up).**

A front contact is placed in series with a parallel pair of B and C front contacts. If A drops, neither branch can energise R. If A is up, either B or C can provide the remaining path.

The complementary statement is that R is down if A is down, **or** if both B and C are down. Changing from energisation conditions to de-energisation conditions changes AND and OR relationships. Checking

the four or eight possible input combinations is often clearer than trying to interpret a long sentence mentally.

## 9.11 Memory: the stick circuit

A momentary push button can energise a relay, but releasing the button opens that path. A **stick circuit** places a front contact of R in parallel with the starting button. Once R has picked, its own contact provides a holding path.

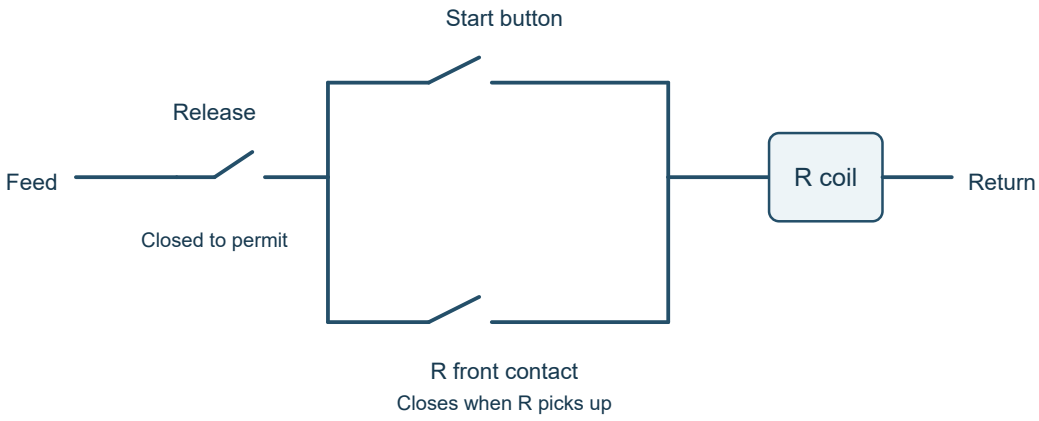



Figure 9.1: A conceptual stick circuit. The release condition must remain in series with both the start and holding paths.

The circuit also needs a way to release R. A release condition in series with both paths removes the holding current when required. Placing the release condition only in the push-button branch would leave the stick branch able to hold R.

This is electrical memory rather than a latched relay mechanism. In the basic example, loss of the supply drops R and opens its stick contact. Restoring the supply alone does not restart it when the starting button is open.

## 9.12 Finding a named relay

The function-name table explains the naming convention. Once a name has been interpreted, use the profile and relay specification to establish its physical position, contact arrangement and characteristics. Knowing what a name means does not establish which plugboard position or terminal is required.

## 9.13 Relays controlling aspects

Aspect selection applies the same logic to a visible railway function: the less restrictive aspect requires its positive conditions. Control state, lamp supply and the lamp-proving indication remain separate. The lineside-signals chapter brings those functions together, including aspect-control relays and lamp proving.

## 9.14 Worked example: the missing release

R starts through a button and holds through its own front contact. A track-clear contact is fitted only in the button branch. Once R is up, the track-clear contact opens, but R stays up. Explain the result.

The holding branch bypasses the track-clear contact. The circuit therefore checks that condition when starting but does not require it while holding. If the requirement is for loss of the track-clear indication to release R, the condition must control the holding path as well. A real safety circuit needs its complete design and failure analysis, beyond this ideal example.

## 9.15 Key points

- The magnetic circuit, residual gap and contact mechanism contribute to the relay's behaviour.
- Double-wound and twin describe different relationships between coils and mechanisms.
- Front and back identify contact states relative to relay operation.
- Series paths express AND; parallel alternatives express OR for energisation.
- A stick circuit changes the path available after the relay operates.
- Function names connect contacts to their coils and equipment.
- An ideal logic diagram alone is not a complete safety design.

# 10. Train detection

**i** Learning objectives

Explain track-circuit feed and receiver functions; relate shunting, ballast leakage and drop-shunt sensitivity; distinguish bonding and rail arrangements; explain clearance boundaries, insulation faults, the traction-return interface and interrupters; and compare track circuits with axle counters; and identify the detection, communication and evaluation stages of AZLM equipment.

## 10.1 What must be established?

Train detection establishes whether a defined section is clear or occupied for signalling purposes. It must deal with stationary vehicles as well as moving trains. Its boundaries and identity must be clear in the design and on the relevant plans.

A track circuit uses the running rails as part of an electrical circuit. An axle counter establishes the occupation state from the counted passage of axles at detection points. Both provide an input to signalling logic, but they obtain their evidence differently.

## 10.2 The basic track circuit

A simple d.c. track circuit has a feed connected across the rails at one end and a track relay across them at the other. With no vehicle present, current flows along one rail, through the relay and back along the other rail. The relay can remain energised.

A conducting wheelset connects the rails with a relatively low resistance. This **shunts** the track circuit, reducing the energy available to the relay so that it drops away. In the basic arrangement, relay up represents the proved-clear state; relay down means that clear is not proved.

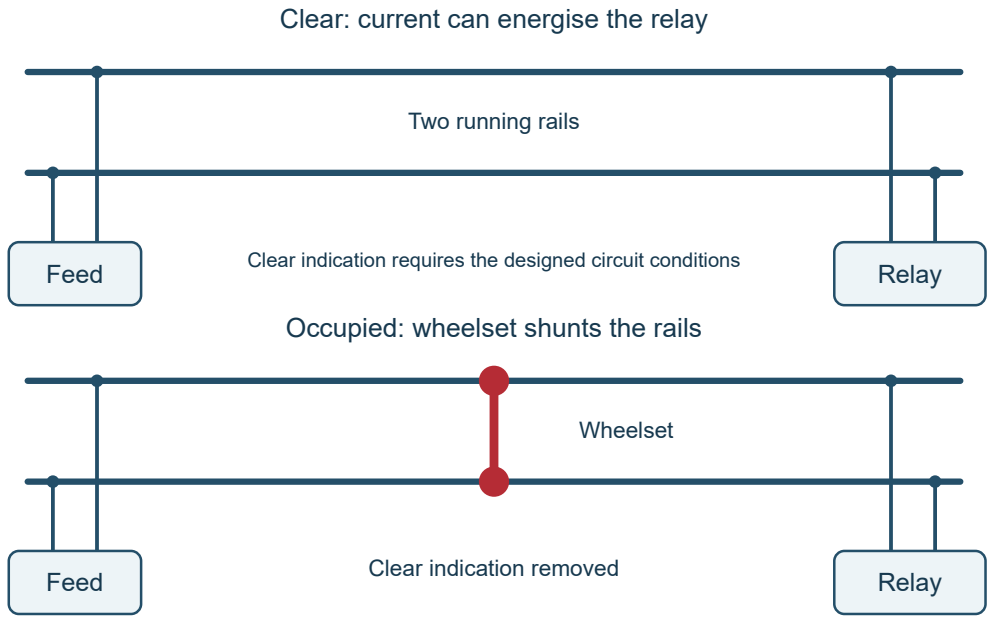



Figure 10.1: A basic track circuit with a wheelset shunt. Circuit limits, protection and detailed equipment are omitted.

Consider a broken connection or loss of feed. Either removes energy from the relay and removes the clear indication. If the opposite convention were used, with an unenergised relay interpreted as clear, loss of supply could falsely report clear while a train was present. The choice of energisation convention is therefore part of the safety principle.

This does not prove that every fault in every track circuit is safe. Unintended feeds, inadequate shunting, failed insulation and equipment faults must also be addressed by the complete design and maintenance arrangements.

## 10.3 Real rails are not an ideal circuit

Current can leak between rails through sleepers, fastenings and ballast. **Ballast resistance** describes this distributed leakage path. Wet or contaminated conditions can increase leakage and reduce the energy reaching the receiver.

The wheel-rail shunt is also imperfect. Contamination, surface films and vehicle conditions can affect contact. A track circuit must distinguish the required clear and occupied states across its specified operating conditions, rather than merely work on a clean demonstration track.

Feed resistance, rail resistance, cable lengths, ballast leakage and receiver characteristics all influence the result. This is why limits and settings are equipment-specific. Increasing a supply to overcome one symptom can compromise another required condition; this book does not provide adjustment instructions.

## 10.4 Feed regulation and the shunting margin

A basic d.c. arrangement includes a **feed or regulating resistance** in the supply path. It limits current when a wheelset shunts the rails and forms part of the electrical balance between the supply, rails, ballast and relay. The resistor may be separate or incorporated in the feed unit.

With the section clear, part of the feed current leaks through the ballast and part reaches the relay. Lower ballast resistance increases that leakage and can reduce the relay input sufficiently to remove the clear indication. The feed setting must preserve the required margins across the specified ballast conditions.

The relay's **pick-up** and **drop-away** characteristics define different transitions. Sufficient input is needed to pick the relay, while the input must fall sufficiently to release it. The required margin has to cover both an unoccupied section and an effective train shunt across the range of specified conditions.

### 10.4.1 What drop shunt means

The **drop-shunt value** is the shunt resistance, applied at the specified test position, that just causes the circuit to change to its occupied state under the test conditions. It expresses sensitivity to a finite shunt; a real wheelset does not provide a mathematically perfect short circuit.

A circuit that requires an exceptionally low shunt resistance before releasing is less sensitive to imperfect wheel-rail contact. A value obtained in one weather or feed condition cannot be assumed to describe every operating condition. Acceptance values, positions, test equipment and adjustment requirements are defined for the particular track circuit.

In a simple regulated d.c. arrangement, increasing feed resistance reduces the rail voltage and can make the relay easier to shunt. Reducing feed resistance can increase the rail voltage and make shunting more difficult. This explains why restoring clear by increasing feed is not an adequate response to poor ballast conditions: the occupied-state margin also matters.

### 10.4.2 Worked example: the weather changes

A teaching track circuit has a satisfactory clear indication in dry conditions. After heavy rain its relay drops even though no vehicle is present. Explain a plausible electrical chain.

Moisture can lower the resistance between the rails through the ballast and support. More current then takes that leakage path, changing the voltage and current available to the relay. If its input falls below the required holding condition, the clear indication is removed. The explanation identifies a possible mechanism, not a diagnosis: feed, connections, rail paths and receiver condition still need to be accounted for.

## 10.5 Relaying the state into the interlocking

The track relay may be in a location case close to the rails while the logic using its information is elsewhere. A repeater circuit carries its state to the required place. A slow-pick track repeater rejects brief pickups associated with intermittent shunting, sometimes called **bobbing**.

Slow pick must be distinguished from slow release. Delaying the acceptance of a newly obtained clear condition serves a different purpose from delaying the removal of the clear indication. The complete timing arrangement is defined by the design.

**Double cutting** places controlling contacts in both feed and return paths. This addresses unintended current paths in external circuits that could otherwise bypass a single controlling contact. It is part of a complete protective arrangement, not a statement that earth faults can be ignored.

## 10.6 Boundaries, joints and bonds

Insulated joints separate the rail circuits where required. Track-circuit identities label the defined sections, while feed and relay-end information helps explain their electrical arrangement.

**Fishplate bonds** provide continuity around ordinary mechanical joints. **Jumper bonds** provide the intended paths through points and crossings. Extra insulated joints can prevent an unwanted short through the turnout rails. A train at every required location must produce the intended shunt, and the effect of a disconnected bond must be considered.

The correct bonding arrangement cannot be inferred from a single-line scheme plan alone. The individual rail paths, joints and connections must be understood from the appropriate detailed information.

## 10.7 Insulation faults in the track structure

An insulated joint can lose electrical separation even while its mechanical connection remains intact. Rail-end burring can bridge the insulating gap; metal fragments or conductive contamination can also create a path across it. The consequence depends on the adjacent circuit arrangements, which is why joint condition must be understood alongside the feeds and polarities on the bonding plan.

Rail fastenings provide another possible leakage path. On reinforced-concrete sleepers, conductive clips, housings and reinforcement can form part of an unwanted connection if the intended insulating components deteriorate. The electrical fault may involve a combination of components rather than one conspicuously broken part.

These mechanisms connect electrical symptoms to the track environment. A relay losing the clear indication after track work can reflect a damaged tail cable, disturbed bond, changed insulation or altered support. The cables chapter covers cable condition and testing; the additional task here is to account for the intended electrical separation of the rails themselves.

## 10.8 Series and parallel bonding

Series bonding arranges the intended feed-to-relay path through the required rails and bonds. A disconnection in that path removes the relay input. The aim is not simply to get current to the relay by the shortest available wire, but to preserve the required detection and failure behaviour throughout the section.

At points and crossings, complete series coverage of every rail path is not always possible. A parallel branch can allow energy to reach the relay while an electrically disconnected part of the layout no longer provides the required train shunt. That is why a bond's existence and purpose matter, rather than just the observation that the track currently works.

Some layouts use additional bonds and special identification to address defined failure risks. Yellow identification can mark bonding requiring particular attention, but colour alone does not establish the complete arrangement. Use the actual bonding plan and equipment requirements.

## 10.9 Double rail, single rail and common rail

These terms describe how rail continuity and insulation are arranged. They must also be interpreted with the traction-return design.

| Arrangement | Distinguishing feature |
|---|---|
| Elementary double-rail d.c. circuit | Both rail paths are insulated at the section boundaries in the arrangement described |
| Single-rail circuit | One rail is continuous for the intended traction-return path; insulated joints divide the signalling rail |
| Common-rail circuit | One common rail is continuous between sections but is not serving as the traction-return rail in this description |

A single-rail circuit and a common-rail circuit can look similar on a small sketch while having different electrical interfaces. Other double-rail arrangements share both rails with traction return through impedance bonds. The name alone does not define every connection.

### 10.9.1 Staggering adjacent circuits

Adjacent polarity-sensitive track circuits can be arranged with opposing feed polarities across their boundary. This **staggering** helps ensure that loss of the intended insulation does not allow a neighbouring feed to sustain a false clear.

The protection depends on the complete feed, receiver and boundary arrangement. Opposing labels on a drawing are not a universal guarantee for every possible joint or bonding failure. Trace the actual rail circuits before explaining the effect of a failed joint.

### 10.9.2 Track-circuit interrupters

Trap or catch points can derail an irregular movement, after which the wheels might no longer shunt the running rails. A **track-circuit interrupter** provides an additional path-breaking function at a defined position. A wheel making the irregular movement breaks the sacrificial element and interrupts the circuit, retaining a restrictive detection state until the required repair and restoration.

It does not detect every derailment anywhere in the section. Its position, mechanical relationship and wiring provide the defined protection at that particular arrangement.

## 10.10 Clearance at points and crossings

A track-circuit boundary used to prove clearance must account for the vehicle body as well as its wheels. A rear wheelset can have passed a joint while the end of the vehicle still projects towards a conflicting route. At a converging junction, the point where the wheelsets have left one section is therefore not automatically the point where the whole vehicle is clear.

The **fouling point** concerns the relationship between the two vehicle paths. The detection boundary and any clearance provision must be located so that the required clear indication has the intended meaning for those paths. Crossing angle, curvature, vehicle overhang and permitted movements influence that relationship; a convenient rail joint alone cannot establish the correct position.

Use the applicable layout and design information to understand the boundary. A clear section can support route release only for the clearance that its arrangement actually proves. Numerical distances and placement rules are design requirements, rather than values to infer from a general illustration.

## 10.11 Track circuits and electric traction

Where running rails carry traction return, they have two electrical roles. Track-circuit boundaries must not simply interrupt the required traction-return path, and traction current must not be accepted as a valid track-clear signal.

Track circuits include d.c., a.c. and dual-immune arrangements. **Immunity** is a designed response to specified interference, including the characteristics of the traction system. It is not a claim that different electrical frequencies can never interact.

An **impedance bond** can provide a path for traction return while presenting the required impedance to track-circuit current. Single-rail arrangements use a different configuration of rail continuity and insulated joints. Selection and bonding depend on the electrical system and the design.

Traction can also couple unwanted voltages into parallel signalling cables. Equipment immunity and protection of people are separate requirements; neither can be established from the nominal working voltage printed on a relay alone.

## 10.12 Receivers, phase and frequency

A basic a.c. track circuit can use a double-element vane relay with a local coil and a control coil. The local coil has its specified local supply, while the control coil receives the track-circuit input. The required frequency and phase relationship creates the operating torque on the vane.

A feed capacitor can provide a required phase relationship and block d.c. in the arrangement concerned. Losing a local supply, altering the phase relationship or losing the rail input can therefore prevent operation even when a voltage is present elsewhere. Voltage magnitude alone does not describe the receiver's full input conditions.

Audio-frequency circuits instead use selective generation and reception at the intended frequency or code. Reed-filter arrangements provide a sharply selective response through mechanical resonance, linking this subject to remote control systems.

An impedance bond connects traction return and signalling through a designed frequency-dependent arrangement. Resonating components can increase the impedance presented to the signalling current. The heavy-current traction path, its balance and the receiver's signal path must all be considered together.

## 10.13 Axle counters

An axle-counter section has detection points at every entrance and exit. An evaluator maintains the count and state. A pair of sensing elements allows the direction of a passing wheel to be established. At a boundary between sections, one movement can count out of one section and into the next.

In the elementary example, 24 axles enter and 24 leave, returning the balance to zero. But a numerical zero alone does not establish a valid clear indication: the equipment must be in a valid state, have the required information from its detection points, and satisfy its designed conditions for clear output.

An axle counter does not depend on a wheelset providing an electrical shunt between the running rails, and does not require rail insulation to define the section. It does not, by itself, provide broken-rail detection. Track circuits must also not be treated as a guarantee of detecting every rail break.

### 10.13.1 Direction and the section boundary

A detection point has a reference direction recorded in the design information. Passing its two sensing positions in one sequence establishes movement in that direction; the opposite sequence establishes the reverse movement. Whether the evaluator adds or subtracts depends on which section boundary the point represents. A detection point is therefore not permanently a count-in point for every movement.

This permits a train to enter a section and return through the same boundary. At a shared boundary between two sections, one passage can reduce the count in the section behind and increase it in the section ahead. The association between the detection point, direction and section is essential to interpreting the count.

An axle counter observes boundary passages. An item placed onto the track inside a section has not necessarily passed a detection point, so it cannot be assumed to appear in the stored count. This limitation matters when relating the detection state to engineering activities.

## 10.14 AZLM axle counters

**AZLM** (also written *Az LM* or *AzLM*) is a multi-section digital axle-counter system. Its trackside detection points communicate with a central **Axle Counter Evaluator (ACE)**, which provides section information to the signalling system. The main equipment groups are:

| Item | Role |
|---|---|
| SK30H rail contact | Two transmitter/receiver coil sets sense passing wheels |
| EAK30H trackside electronics | Connect the rail contact to the communication system |

| Item | Role |
|---|---|
| Communication link | Carry detection-point information to the evaluator |
| ACE | Evaluate the information for the configured sections |
| Power Data Coupling Unit (PDCU), where used | Couple power and data onto the line pair, or provide the specified communication interface |

The ACE includes processing, power-supply and input/output boards. In a two-out-of-two arrangement, two processing channels must produce the required agreement. Serial input/output receives detection-point information; parallel input/output provides section information. CAMTECH's AZLM equipment description shows these equipment relationships.

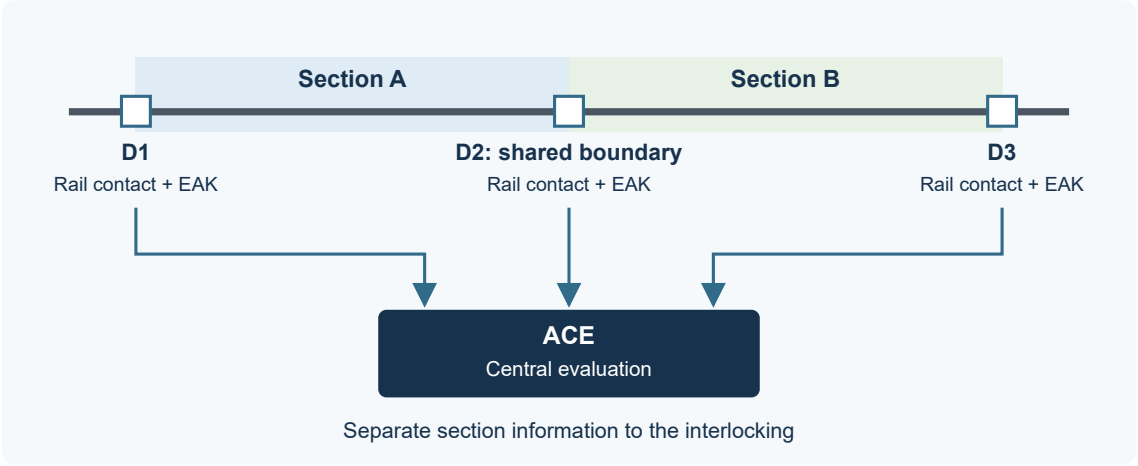



Figure 10.2: Three detection points define two sections, with a shared boundary at D2. The lines below show information flow to the ACE, not a cable wiring arrangement.

### 10.14.1 From a wheel to a section state

The transmitter and receiver coils at the rail contact are magnetically coupled. A passing wheel changes the received signal. The two sensing positions provide a sequence from which wheel passage and direction can be established; the two channels must be interpreted together.

In AZLM, the trackside electronics carry out counting supervision and generate digital messages for the ACE. The ZP30H detection-point arrangement uses two independent microcontrollers, with supervision from the evaluator. Its communication uses coded telegrams over the specified link. A voltage present on the line pair alone cannot establish that valid count information is reaching the intended evaluator.

The ACE processes the received information through its vital computer system and produces occupancy information through the configured output interface. The detection point, message path, processing and output are distinct stages to account for when studying a failure. Network Rail's AZLM description, Appendix B.3, sets out this architecture.

### 10.14.2 Worked example: a train crosses a shared boundary

In Figure 10.2, sections A and B are initially valid and clear. A train of 12 axles enters A at D1. The train then crosses D2 towards B. After five axles have passed D2, the teaching count is seven in A and five in B: both sections are occupied.

After all 12 axles have crossed D2, A can return to clear if all its validity conditions are satisfied; B still contains 12 axles. It can return to clear after the corresponding departures at D3 and the required valid processing. The shared detection point supports two different section balances. It does not make the two sections one continuous count store.

### 10.14.3 Read indications as evidence

Keep three questions separate: is the detection-point equipment functioning, is its information being received correctly, and what section state has the evaluator established? A healthy local indication does not, by itself, answer all three. Similarly, a group of affected sections can suggest a shared boundary, communication or supply dependency without proving which component is faulty.

Identify the installation, detection-point identities, associated sections, equipment versions and applicable diagnostic information. Record relevant indications and the circumstances before any authorised action that could clear retained evidence. Rail-head alignment, mounting, cables, electronics, communication and power belong to the complete detection chain. The cable and power-supply chapters explain the electrical paths without repeating their test principles here.

### 10.14.4 Resetting and restoration

A **reset** changes the evaluator's counting or section state under its defined conditions. **Restoration** returns the detection function to operational use through the required controlled arrangements. Neither finding a physically empty-looking section nor obtaining a zero display establishes all the evidence needed for restoration.

The permitted reset method and its prerequisites depend on the installed system and application. Follow the relevant controlled instructions for AZLM; a plunger sequence, relay arrangement or reset inhibit from another axle-counter family cannot be assumed to apply. This chapter explains the distinction, rather than prescribing a reset or restoration sequence.

## 10.15 A methodical account of a failure

Begin with the expected state and the observed symptom. Establish the section identity, feed, receiver and any repeaters before explaining why the displayed state differs from expectation. A panel indication, local relay state and rail input are related evidence, but they refer to different points in the chain.

Group possible causes by function: energy generation, the feed path, rail continuity, unwanted shunts or leakage, receiver operation, and transmission of the resulting state. This keeps a missing indication from being treated automatically as a failed track relay. Compare the symptoms with shared supplies and recent work as well as with the individual circuit.

Record the conditions associated with observations, including weather and whether the symptom is intermittent. Measurements must be interpreted against the circuit type and authorised test arrangements. A structured explanation narrows the uncertainty; it does not justify raising the feed or bypassing a proving condition merely to obtain a clear indication.

## 10.16 Worked example: an empty section shown occupied

A section appears physically empty but has no clear indication. Is that necessarily a wrong-side failure?

No. If loss of feed, a connection fault or invalid evaluator information removes the clear indication, the signalling system may correctly take the restrictive state. The physical observation and the equipment state are different evidence. The response should be interpreted from the system design, without treating appearance alone as authority to restore a clear indication.

## 10.17 Key points

- The basic track circuit requires positive energy for the clear indication.
- Shunting and ballast leakage affect the electrical balance differently.
- Feed regulation must preserve both the clear-state and occupied-state margins.
- Rail configuration and bond layout determine the consequences of a disconnection.
- Bonds, joints and traction return are integral to the design.
- Detection used to prove clearance must account for the vehicle body beyond its wheelsets.
- Axle counters depend on valid counting and system state.
- AZLM separates trackside detection, digital communication, central evaluation and section outputs.
- A restrictive output can represent a fault as well as a train.

# 11. Interlocking

**i** Learning objectives

Follow route setting from availability to signal clearance; distinguish the main forms of locking; explain cancellation and release; and identify how control tables record the requirements.

## 11.1 From a command to a protected movement

Interlocking prevents incompatible routes, controls when points may move and determines when a signal may clear. It must maintain the required protection while a train approaches and traverses the route, and release it only under the specified conditions.

The entrance-exit interface simplifies entering the signaller's route command. It does not collapse the engineering process into one event. **Route commanded**, **route available**, **route locked** and **signal cleared** describe different stages.

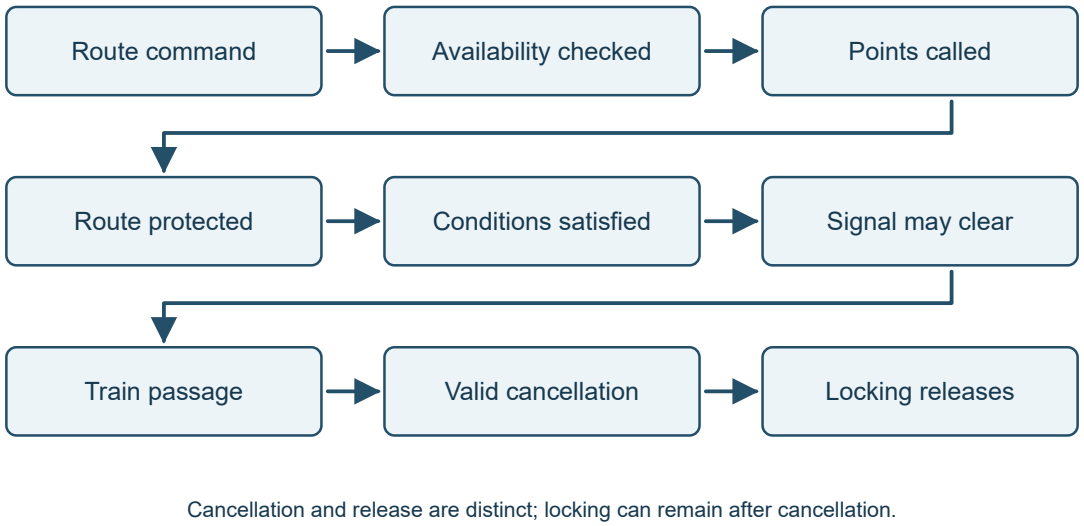



Figure 11.1: A simplified route-setting and release lifecycle. Each transition is subject to its designed conditions.

## 11.2 Availability before movement

First establish that the commanded route exists and that its required points are available. A point already in the required position may be available; a point locked in the opposite position is not. The availability check prevents unnecessary point movements for a route that cannot be established.

The system then calls the required points and establishes route locking. Point detection supplies the required indications to the signal controls. Availability and detection answer different questions: "may this point be used as required?" and "is the required state proved?"

## 11.3 Route locking and conflicting movements

Route locking prevents the points being changed in a way that would destroy an established movement. Locking functions are associated with the successive train-detection sections of the route and overlap.

Some conflicting routes require a common point in opposite positions. Locking the point for the first route prevents availability for the second. Other conflicts cannot be excluded that way. In particular, opposing routes can use the same point positions while authorising trains towards one another. Directional route conditions must exclude that conflict.

White route indications and red occupation indications help the signaller follow the state. Their detailed meaning depends on the interface. They are information about the underlying functions, rather than the locking mechanism itself.

## 11.4 Conditions for clearing the signal

For an invented main route from S1 to S3 over points P1, consider the simplified requirement:

**Route set and locked AND P1 detected reverse AND T1 clear AND T2 clear AND overlap O1 clear AND the required S3 proving satisfied.**

Each item has a purpose:

| Condition | What it establishes in this example |
|---|---|
| Route set and locked | The correct route exists and its required protection is established |
| P1 detected reverse | The required point state is proved |
| T1 and T2 clear | The route sections required clear are proved clear |
| O1 clear | The required overlap section is proved clear |
| S3 proving | The specified signal-in-advance proving is satisfied |

This is a learning expression, not a complete control table. Flank conditions, aspect sequence, approach controls and other requirements can also apply. If any required term is false, the expression does not permit clearance.

## 11.5 Approach control and approach locking

These terms sound alike but serve different purposes.

**Approach control** regulates when a signal clears as the train approaches. For a junction with approach control from red, a signal is held at danger until an approach condition is satisfied so that the train has reduced speed before taking a divergence. Depending on the design, occupation, timing and aspect conditions can be involved.

**Approach locking** maintains route protection after a signal has been cleared and is then replaced. A driver may already have acted on the proceed aspect and may be unable to stop before the signal. Replacing it to danger does not mean that the route locking has released or that the route is immediately free to alter.

A release may depend on a look-back assessment, a timing condition, passage of the train or another designed sequence. No generic delay from an introductory example should be applied as a site instruction.

## 11.6 Train-operated cancellation and sectional release

**Train Operated Route Release (TORR)** uses a defined sequence of train-detection operations associated with passage of the train. One occupied input alone is insufficient: a detection failure could otherwise mimic the event used to cancel a route.

Cancellation and release must be distinguished. A route can be cancelled while locking remains in force. Once the release conditions are met, **sectional release** removes locking behind the train in the specified sequence. A section ahead of the train must not become available merely because a preceding section has cleared.

A simplified learning sequence is:

1. The signal clears with the route protected.
2. The train passes the signal and the signal is replaced.
3. A valid passage sequence cancels the route command where TORR is provided.
4. Locking releases progressively as the required sections clear behind the train.
5. Remaining overlap or other locking releases under its own specified conditions.

Different layouts need different release conditions. The governing requirements are recorded for the actual route.

## 11.7 Point locking has several causes

| Type | Purpose |
|---|---|
| Route locking | Preserve an established route or protected arrangement |
| Dead locking | Prevent point movement when the associated train-detection section is occupied |
| Individual-control or key locking | Hold points in a selected position through the control arrangement |
| Time-of-operation locking | Address movement timing where a train could reach points before a newly initiated movement finishes |

Dead locking and route locking can apply simultaneously. Waiting until the point section becomes occupied may be insufficient where the points are close to its boundary. Locking must address the train's approach and the equipment's movement time, rather than just its eventual position.

## 11.8 Flank protection

A route can need protection from equipment outside its direct line of travel. Points in a siding or on an adjacent connection may be positioned and locked to prevent an unwanted movement reaching the route. This is **flank protection**.

The precise detection indications required depend on the arrangement. Do not assume that every flank point has the same detection requirements as a point traversed by the train. Read the specified controls, including any additional track-clear conditions associated with alternative protection.

## 11.9 Overlaps and alternative paths

An overlap can include several sections and points. Trailing points can require a particular locked position. Facing points can provide alternative overlap paths if the design permits a **swinging overlap**.

A swinging overlap is a controlled change between valid protected arrangements. It does not mean that the points are unrestricted. Availability, detection, conflicts, timing and train occupation must all be addressed as required. Overlap release may also differ from route release, for example where a train stops at a platform and a later movement needs the points beyond it.

## 11.10 Shunt routes and ground frames

A shunt route may not require every section up to its destination to be clear, because the movement can be intended to approach standing vehicles. This does not make its point detection or route protection optional. Each movement has its own requirements.

A ground frame provides local control of a limited group of equipment. A release from the controlling signaller enables the local operation under the specified protective conditions. Restoring the frame and its equipment to the required state is part of handing control back. A release is an engineered interface between two control arrangements, not merely permission conveyed by an indication lamp.

## 11.11 Control tables

Control tables record the requirements for routes, points and other signalling functions. Signal-route entries include setting conditions, point and track proving, aspect controls, replacement, approach locking and train-operated release. Point entries describe movement restrictions, route locking, releases and overlap conditions.

Separate signal and point sheets organise the requirements by purpose. Start by identifying what a row or column governs, then relate each entry to the layout and the other tables. Do not infer a complete requirement from a fragment of circuit or software data.

## 11.12 Worked example: cancelled but still locked

A signaller cancels a route just after a train has seen its proceed aspect. The signal returns to danger, but the points remain locked. Is the locking necessarily faulty?

No. Approach locking may be preserving the path for a train that cannot stop in time. If the train enters the route, its occupation and subsequent release sequence may continue to hold the necessary locking. The visible replacement of the signal is only one event in the complete process.

## 11.13 Key points

- Availability, locking, detection and aspect control are different stages.
- Opposing routes can conflict without requiring opposite point positions.
- Approach control governs clearance; approach locking protects an approach after clearance.
- Cancellation does not automatically release locking.
- Control tables define the requirements for the actual arrangement.

# 12. Solid State Interlocking

**i** Learning objectives

Identify the principal SSI modules and interfaces, distinguish safety from availability, explain fixed program and application data, and follow a control and its feedback through the trackside data link.

## 12.1 What SSI means here

This chapter uses **SSI** for the British Rail-developed Solid State Interlocking system. It is a particular system architecture, not a name for every electronic interlocking.

SSI implements the interlocking functions introduced in the previous chapter using processors, application data and trackside interfaces. The change of technology does not remove the need to prove points, exclude conflicts and retain locking for a train.

## 12.2 The interlocking cubicle

The **Multiprocessor Modules (MPMs)** carry out the interlocking processing. Three modules compare their results and internal information. In the two-out-of-three arrangement, a disagreeing module can be isolated while the two agreeing modules continue. If the remaining pair disagree, continued safe agreement is no longer established and the system shuts down as designed.

This combines two purposes. Comparison supports the detection of an erroneous result. Having three modules also permits continued operation after one has been isolated. Three copies do not make the application data correct: all three could process the same incorrect requirement consistently. Safety therefore also depends on the program, data, engineering process and interfaces.

| Item | Principal role |
|---|---|
| MPM | Interlocking processing and associated communications |
| Panel Processor Module (PPM) | Interface to the signaller's control equipment |
| Diagnostic module | Monitor system behaviour and provide diagnostic information |
| Technician's terminal | Display diagnostics and provide specified technician facilities |
| Data Link Module (DLM) | Interface to the relevant data-link path |
| Trackside Functional Module (TFM) | Interface between data telegrams and local railway equipment |

## 12.3 Fixed program and application data

The **fixed program** provides the common processing functions. The **application data** defines the particular railway: equipment identities, route conditions, point controls, release sequences and other required relationships.

The distinction makes a reusable system possible, but application data remains safety-critical. It must represent the approved requirements accurately. Data preparation at a design workstation is followed by compilation and simulation before use. Successful compilation establishes that data can be translated; it does not prove that every railway requirement has been represented correctly.

A useful learning chain is **layout and requirements → application data → compiled representation → tested behaviour**. Each transition can introduce an error and needs the appropriate assurance.

## 12.4 The signaller and technician interfaces

The PPM links SSI with a hard panel or electronic control interface. The PPMs are duplicated for availability. That is different from the triplicated interlocking comparison function. A control interface commands actions; the interlocking remains responsible for enforcing its conditions.

The technician's terminal and recording facilities make internal state and events accessible. Time-related records help reconstruct the order of commands, occupation changes, detection changes and outputs. Their interpretation depends on what was actually recorded, its timing and the corresponding equipment indications.

Specified technician controls can impose additional restrictions, such as holding a signal at danger. They do not provide a general means of bypassing normal interlocking conditions. Use of those facilities requires the relevant authority and instructions.

## 12.5 Trackside functional modules

A TFM receives a control telegram and returns indication information. The SSI trackside arrangement supports addresses 1-63 on a trackside system, with actual capacity and spare allowance considered in the scheme design. Addresses identify which module a telegram concerns; the correct association between address, wiring and application data is essential.

Duplicated internal channels within a TFM compare their operation. If agreement is lost, the module is taken out of service as designed. This is distinct from the three-MPM arrangement and has a different effect on availability: a TFM affects its associated functions, while failure of the central interlocking can affect the whole area.

### 12.5.1 Signal module

The signal TFM provides controlled outputs and indication inputs. Outputs can drive appropriate signal loads and other specified equipment. Current-sensing paths return evidence about the connected loads. Input circuits include checking intended to distinguish valid locally supplied input information from a spurious voltage.

**Red retaining** is a special arrangement supporting the most restrictive aspect when a module is disabled in the specified way. It depends on the designed circuit and available power. It is not a claim that every loss of a TFM or supply must leave a visible red.

### 12.5.2 Points module

The points TFM is designed around the outputs and inputs needed for point operation and detection. Its direct outputs suit clamp-lock equipment. Electric machines require the specified relay interfaces. A shared word such as "output" does not imply that every load can be connected directly.

## 12.6 Data links and long-distance interfaces

Duplicated trackside data links carry controls and indications. DLMs interface the electrical paths. Repeaters extend the supported baseband arrangement within its design limits. For longer transport through a telecommunications bearer, **Long Distance Terminals (LDTs)** provide the required interface.

An internal data link provides communication between interlockings. Do not confuse that boundary information with the trackside link to TFMs or the panel interface to the signaller.

Duplicating the links improves availability only to the extent that common failures have been addressed. Two paths in one vulnerable cable route can both be lost together. The physical routing and the equipment shared by them therefore matter as well as their logical names.

## 12.7 Other electronic interlockings

Other electronic interlockings, including Ebilock, VPI and WESTRACE, implement electronic safety in different ways. These include comparison of diverse processing, coded or checked logical processing and

checked input/output architectures. The general lesson is that a system's safety case cannot be inferred from the number of processors alone.

Equipment-specific architecture, diagnostics and permitted actions must be learnt for the actual system. SSI terminology should not be assumed to describe another manufacturer's modules directly.

## 12.8 Worked example: duplication and a cable cut

A and B links are both working, but they pass through the same damaged section of route. Both fail together. Why has duplication not preserved communication?

The paths were logically separate but shared a physical failure exposure. Availability depends on independence from the failures being considered, not merely on providing two interfaces. Route diversity and common equipment must be considered together.

## 12.9 Key points

- SSI implements signalling requirements using processing, data and local interfaces.
- Processor agreement and correct application data are separate requirements.
- Duplication and triplication serve defined safety and availability purposes.
- TFMs translate between telegrams and local equipment.
- Diagnostics provide evidence; they do not replace the complete system documentation.

# 13. Remote control systems

> **i** Learning objectives
>
> Explain the purpose of multiplexing, distinguish time division from frequency division, identify the safety boundary in a remote-control arrangement, and describe the principle of reed equipment.

## 13.1 Why transmit many functions together?

Providing a dedicated pair of long-distance wires for every control and indication becomes costly as an area grows. Consider 200 controls and 500 indications over 30 km. At two conductors per function, that is 1,400 conductors and 42,000 conductor-kilometres.

Conductor-kilometres describe the total length of individual conductors. They do not mean that the cable route is 42,000 km long. The route remains 30 km in the example. Keeping those quantities distinct makes the engineering and cost argument clear.

**Multiplexing** shares a transmission path between functions. The functions remain identifiable by a rule, such as an allocated time slot or frequency channel.

## 13.2 Time Division Multiplex

In **Time Division Multiplex (TDM)**, each function occupies an allocated part of a repeated sequence. Imagine matched switches at the sending and receiving ends stepping through inputs and outputs in the same order. Input 1 is conveyed to output 1, then input 2 to output 2, and so on. The rotating switches are an analogy; the equipment performs the function electronically.

Synchronisation keeps the two ends aligned. If alignment is lost, the receiver must not simply treat one function's information as another's without the system's defined response.

A panel-to-interlocking TDM system carries signaller controls to a satellite relay interlocking and brings indications back. The local interlocking enforces its safety conditions. This allows the control transmission to have a different safety role from a direct vital proving circuit.

> **!** Locate the safety boundary
>
> A remote route command and a vital point-detection input are different kinds of information. Whether a transmission path must itself be vital depends on the function and the complete architecture. Panel-to-interlocking TDM can be non-vital because the interlocking checks the commands. That does not make all signalling transmissions non-vital.

## 13.3 Reconstructing indications

An indication that flashes at the panel need not be transmitted as a series of timed on/off samples. A scan could distort that pattern. The system can send state information such as "flashing required" or "steady required", then generating the actual presentation locally.

This separates the information from its display. A panel processor can reconstruct several indications from the transmitted state, reducing the number of transmitted functions. The display must still be associated with the correct equipment and have a defined response to lost or invalid information.

## 13.4 Frequency Division Multiplex

In **Frequency Division Multiplex (FDM)**, functions share the pair by using different frequency channels. Function A can transmit at frequency f1, B at f2 and C at f3. A receiver for A responds to the intended f1 signal and controls its associated follower relay.

| Feature | TDM example | FDM example |
|---|---|---|
| How a function is distinguished | Position in a timed sequence | Allocated frequency channel |
| Main coordination need | Synchronisation and correct message association | Stable generation, selective reception and compatible channel allocation |
| Typical application | Block transfer between control centre and satellite interlocking | Collection or distribution of trackside functions |

Several frequencies can be present together on the pair. Their separation must be engineered: filtering, harmonics, interference and equipment faults matter. “Different frequencies” alone is not a complete argument for independence.

## 13.5 Reed equipment

A reed is a mechanical resonator whose vibration is determined by its physical properties. A reed filter uses two reed assemblies on a coupled frame. An incoming electrical signal drives one reed; mechanical vibration is transferred to the other; its coil provides an electrical output when the intended resonance is established.

In the receiver, this selective transfer supports acceptance of the assigned frequency. In the transmitter, the reed filter forms part of the oscillator’s frequency-control arrangement. The intended result is a stable, recognisable channel rather than a frequency that can drift into another function’s channel.

Reed equipment has separate filter and electronic units, coding to prevent incorrect substitution, and follower relays matched to the receiver. Vital and non-vital versions have different roles. Physical similarity is not sufficient evidence that equipment has the required frequency, safety role or output characteristics.

## 13.6 The line is part of the system

The transmitter’s connection must not interrupt every other function when one unit is removed. The line arrangement retains continuity, with receivers connected across the pair. Line levels and impedance also affect the whole circuit.

Amplifiers maintain signal level where required. In a vital frequency-based arrangement, they must preserve the characteristics on which safe recognition depends. Isolation addresses electrical conditions along the railway, including induction from traction. Equipment selection and positioning follow the actual design, rather than a generic distance rule.

A **simplex** path carries information in one direction. A **half-duplex** path uses both directions at different times. A **full-duplex** path supports both directions simultaneously. These terms describe direction and timing of communication, not whether the signalling function is safe.

## 13.7 Worked example: identical symptoms from a shared path

Six distributed indications disappear together. Each has its own local source contact, but all use one FDM pair and a common receiving supply. What does the pattern suggest?

It identifies shared dependencies worth comparing with the system records: the line pair, common power, amplification or receiving equipment. It does not prove which item failed. Independent local faults remain possible, but a simultaneous group change is evidence about the system structure.

The useful reasoning step is to map the affected functions to what they share before interpreting each as an unrelated equipment failure.

## 13.8 Key points

- Multiplexing reduces dedicated transmission wiring by sharing a path.
- TDM separates functions in time; FDM separates them by frequency.
- Safety requirements depend on the function carried and its place in the architecture.
- Reed systems rely on controlled resonance and selective reception.
- Shared line and power arrangements can produce groups of symptoms.

# 14. Telecommunications

**i** Learning objectives

Describe the main railway voice and data services, explain the purpose of a telephone concentrator and route diversity, relate digital bearer allocation to signalling services, and identify the telecommunications interface to SSI.

## 14.1 Services and bearers

Telecommunications carries voice and data between the people and equipment operating the railway. A useful distinction is between a **service**, such as a signaller-driver call or an SSI data connection, and its **bearer**, such as a copper circuit, optical transmission path or radio link.

Several services may share a bearer. A service can also use more than one bearer to improve availability. To understand a failure, identify both the user-facing function and the infrastructure that carries it.

Telecommunications includes lineside telephones, concentrators, SSI links, transmission systems, passenger information and radio. Their equipment differs, but common questions recur: who is connected, how is the circuit identified, what information is carried, what delay is acceptable, and what happens when a path fails?

## 14.2 Lineside telephones

A **signal post telephone (SPT)** provides a direct connection associated with a particular signal and controlling location. Other direct lines serve level crossings, adjacent control locations and operational interfaces. Dial telephones provide access to a telephone network, while maintenance plug points support communications between staff at equipment locations.

Identity is essential. The person answering must know the circuit involved, and the caller must identify their location and purpose through the applicable communication arrangements. A technically clear voice path to the wrong person or location does not meet the operational need.

Outdoor telephones also need suitable environmental protection, labels and physical robustness. Their lines and housings are part of the service, alongside the switching equipment at the control centre.

## 14.3 Telephone concentrators

A concentrator brings many dedicated operational circuits to the signaller's console. Its functions include circuit identity, controlled call selection, prevention of unintended overhearing or interruption, and clear indication of which caller is connected.

Call parking lets a caller remain associated with the system while the signaller deals with another call. It is different from joining callers together. Distinctive indications identify urgent or special calls. Recording provides evidence of communications, with the associated need to preserve the relevant record when required.

The engineering objective is more than making speech audible. A dependable operational service must deliver speech to the intended party and make its identity unambiguous.

## 14.4 Diverse routing

If adjacent lineside phones all depend on the same vulnerable cable path, one cut can remove several means of communication together. Alternating route directions and transmission paths can reduce the effect of a single fault on neighbouring facilities.

**Diversity** must be assessed physically. Separate circuit numbers do not prove separate cables, ducts, power supplies or transmission nodes. The same point applies to duplicated signalling data links.

Remote concentrators can combine low-use local circuits before sending them over a longer path. This reduces dedicated long-distance wiring while preserving the required call-control functions. The network arrangement must still address power, identity, capacity and failure behaviour.

## 14.5 SSI long-distance links

An SSI baseband link uses the appropriate DLM interfaces and trackside cabling. An LDT adapts it for transport through a telecommunications bearer, with a corresponding interface at the remote end.

The bearer must meet the signalling service's requirements, including timing. A path can be electrically intact and carry data, yet fail the service if delay or the relative arrival of duplicated messages is outside the permitted limits.

For learning, separate four checks:

1. **Continuity:** is a path available?
2. **Integrity:** is the required information correctly conveyed?
3. **Timing:** does it arrive within the specified constraints?
4. **Independence:** what failures can affect both A and B paths?

The permitted delay and message-timing limits depend on the SSI/LDT arrangement. They must be established from the applicable interface and design documentation.

## 14.6 Digital bearer allocation

Digital transmission equipment can carry several operational voice and signalling data services over one bearer. PCM channel and aggregate interface rates describe available transmission capacity; the circuit allocation and interface records establish which service reaches which destination.

For a signalling link, identify its allocated channel, end interfaces and required timing and integrity. A healthy aggregate bearer or a familiar interface rate does not establish that the correct signalling service is connected end to end. This is particularly important after transmission alterations or a change of route.

## 14.7 Multiplexing and transmission hierarchy

TDM shares transmission capacity through timed channel positions. Transmission systems include **Plesiochronous Digital Hierarchy (PDH)**, with tributary clocks operating within tolerances, and **Synchronous Digital Hierarchy (SDH)**, with a coordinated timing structure supporting access to channels at intermediate nodes.

A **drop-and-insert** point lets channels be taken from or added to a transmission system without terminating the entire service path. Digital regeneration reconstructs a valid signal at a stage, rather than simply amplifying everything present, including noise. It still depends on a receivable input and a correctly operating system.

An optical fibre carries signals as light and removes a conductive connection along that fibre. It does not remove the need for powered terminal equipment, correctly engineered interfaces and a resilient physical route.

## 14.8 Passenger information and radio

Customer information systems include public address, recorded announcements, displays and clocks. Automatic updates depend on the correct association between service information, location and time.

Radio provides a mobile communications path. Coverage, selective calling, emergency calls and association between a train and the relevant control area are important radio-system functions. These are examples of system functions, rather than an inventory of equipment at a particular site.

Voice communications, a data bearer and the safety system using that data are distinct functions. Radio coverage alone does not establish a safe movement authority. The complete train-control arrangement must define and assure that authority and its supervision.

## 14.9 Worked example: connected but unusable

Both ends of a signalling data service report a connection, but valid messages are rejected after a transmission route change. Why is "the link is up" insufficient evidence that the service is satisfactory?

Connection status says little about the message integrity, latency or relationship between duplicated paths. A new route can change those characteristics. The service must be assessed against its end-to-end requirements, using the evidence and responsibilities defined for that interface.

## 14.10 Key points

- Services and bearers are different layers of the system.
- Operational speech needs correct identity as well as audibility.
- Route diversity addresses common physical failures.
- Data services can have strict timing as well as continuity requirements.
- Channel allocation and interface records connect a signalling service to its bearer.

# 15. Signalling power supplies

**i** Learning objectives

Relate supply condition to signalling load and duty; interpret battery standby and charging arrangements; identify the roles of local conversion and isolation; explain alternative supplies and fuse discrimination; and follow a supply from its source to the load.

## 15.1 Start with the load

Signalling equipment needs energy as well as control information. A relay coil, a signal lamp and a point motor have different voltage, current and duty requirements. A supply must suit its connected load and remain adequate during the conditions for which the installation is designed.

A continuously energised relay imposes a steady demand. A point movement imposes a relatively large intermittent demand. A signal can change which lamp is energised while continuing to require a dependable supply. These patterns influence the choice of supply, protection, distribution and standby arrangements.

The main devices introduced here perform different jobs:

| Device | Main function |
|---|---|
| Cell or battery | Produce electrical energy through chemical action |
| Transformer | Transfer a.c. energy between windings and provide the designed voltage ratio and isolation |
| Rectifier | Convert a.c. into unidirectional current |
| Charger | Supply the controlled charging conditions needed by a rechargeable battery |
| Distribution and protection | Convey energy to the loads and control the consequences of faults |

A transformer-rectifier can supply d.c. equipment or form part of a charging system. Those functions do not make every transformer-rectifier a complete battery charger.

## 15.2 Batteries in signalling supplies

A rechargeable signalling battery provides reserve when its normal supply is unavailable and, in some arrangements, supports intermittent equipment demands. Establish the battery type, nominal voltage, cell arrangement, charger and loads from the supply records. The way the cells or strings are connected affects the available voltage, capacity and charging requirements.

Battery identity matters when interpreting readings or selecting replacement equipment. A series string depends on every cell and intercell connection; a weak cell or resistive connection can limit the performance of the complete supply. Where parallel strings are provided, their intended current sharing and protection are part of the design.

### 15.2.1 Primary cells: background

A **primary cell** supplies energy through a chemical reaction that is not intended to be reversed by charging; once exhausted it is replaced. Zinc-carbon dry cells and wet primary types have been used in signalling where a local self-contained supply was needed. They are useful background knowledge for

interpreting older drawings, but the practical battery coverage in this book concentrates on rechargeable supplies.

## 15.3 Battery voltage under signalling load

A satisfactory unloaded voltage does not prove that a battery can support its signalling duty. Internal resistance, cell condition and intercell connections can produce a marked voltage fall during a point movement or other demand. Distinguish a fall at the battery terminals from the additional drop between the battery and the equipment.

Record the load condition with the voltage. A reading taken while the battery is on charge, supplying its ordinary relay load or supporting a point movement describes a different condition. Compare the evidence with the applicable battery and equipment requirements.

## 15.4 Standby capacity and discharge rate

The stated Ah capacity must be read with its rated discharge conditions and end voltage. The important signalling question is whether the battery can maintain its defined loads for the required standby period, including any intermittent demands. Temperature, age, state of charge and discharge rate affect usable reserve.

A battery can cease to support a relay, lamp or point machine before all its nominal capacity has been used, because the equipment's minimum input condition has been reached. Capacity, charger indication and open-circuit voltage are therefore different evidence about supply performance.

## 15.5 Lead-acid cells

A lead-acid secondary cell has positive and negative plates separated within an acid-resistant container. Its electrolyte is dilute sulphuric acid. The chemical condition of the plates and electrolyte changes as the cell charges and discharges.

For accessible flooded cells, electrolyte **specific gravity (SG)** can provide evidence of charge state when interpreted with the cell type, temperature and condition. A hydrometer measures liquid density relative to water; SG is dimensionless, not a voltage reading. Designs with sealed or otherwise inaccessible electrolyte require their own assessment methods.

Prolonged discharge can allow persistent sulphation to develop, reducing usable capacity and increasing internal resistance. A battery's condition therefore cannot be established solely from its external appearance or the charger being switched on.

## 15.6 Rechargeable alkaline cells

Rechargeable alkaline cells use an alkaline electrolyte, commonly potassium hydroxide in the types considered here. Their construction, voltage characteristics and charging requirements differ from lead-acid cells. Nickel-cadmium and nickel-iron identify different electrode systems and must not be treated as interchangeable names.

Electrolyte specific gravity does not track state of charge in the same way as in a lead-acid cell. A familiar hydrometer reading must therefore not be given the same interpretation across different chemistries. Battery identity is essential before interpreting test results or selecting charging equipment.

## 15.7 Charging arrangements

**Initial charging** prepares a cell or battery for service where its supply condition requires it. **Recharging** restores charge after discharge. Neither has a universal current or duration applicable to all batteries.

A **trickle-charging** arrangement supplies a relatively small charging current, associated with replenishing charge between intermittent demands. The average load and charging capability must be compatible; a short operating cycle can create a high peak demand while drawing relatively little average current.

A **float-charging** arrangement maintains the battery at its specified standby charging condition while the supply supports the normal load. On loss of the normal supply, the battery can support the circuit for its designed duty. Correct voltage control and current capability remain necessary.

A regulated **constant-voltage charger** holds the designed output characteristic while its current changes with battery and load conditions, within its rating. Constant voltage does not mean constant charging current or unlimited output current.

Undercharging leaves insufficient reserve and can damage some chemistries. Excessive charging can cause heating, electrolyte loss and gas evolution. The charging system must match the battery, its application and the manufacturer's requirements.

## 15.8 Local transformers

Signalling distribution can use a higher a.c. voltage with local transformers providing the supplies needed by lamps, track circuits and other equipment. Identify the primary supply, secondary circuit, rating and selected tapping from the actual installation. Conversion, isolation and the condition of the downstream circuit must all be accounted for.

Transformer tappings provide the designed output choices. Selecting a different tapping is not a general remedy for low voltage at equipment: first distinguish the transformer output from voltage lost in cables, contacts and terminations. A satisfactory no-load secondary voltage does not establish that the circuit remains adequate under its signalling load.

## 15.9 Isolation and earth-free circuits

Isolation can separate a signalling supply from the earthing arrangement of its upstream source. In an **earth-free** circuit, an unintended connection to earth can remain hidden until another fault creates an unwanted path. The circuit's protective and monitoring arrangements are therefore part of the supply system.

An isolating transformer does not make every downstream part harmless to touch. A voltage still exists between its output conductors, and other sources or induced voltages can be present. The actual circuit and isolation arrangements govern the work.

## 15.10 Transformer-rectifiers

A signalling transformer-rectifier combines local voltage conversion with a d.c. output for its specified equipment. Establish whether the installation also provides smoothing, regulation or battery charging; those functions cannot be assumed from a transformer-rectifier label.

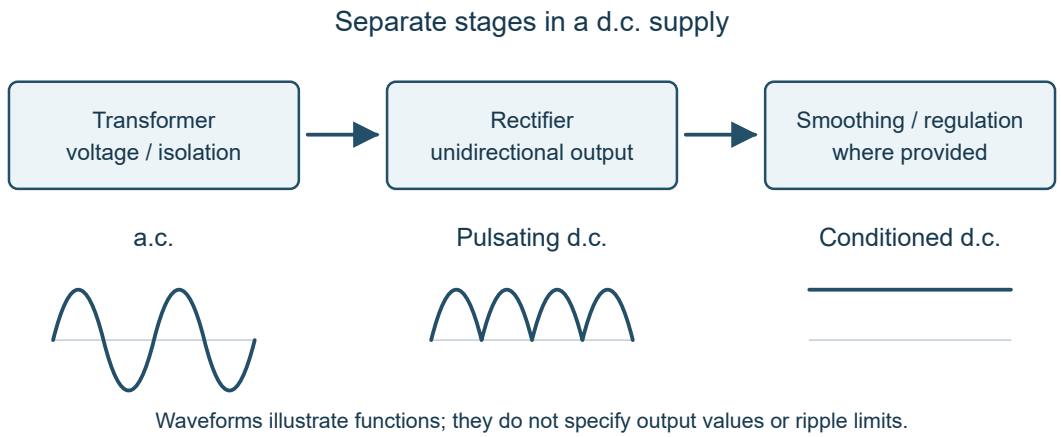



Figure 15.1: Voltage conversion, rectification and any smoothing or regulation are distinct stages.

A missing or defective stage can produce different symptoms: no output, low output, excessive ripple or inadequate performance under load. A label stating d.c. does not mean that the output has no ripple, and a correct no-load reading does not prove rated load capability.

## 15.11 Follow the distribution path

A typical study path is **incoming supply → distribution and protection → local conversion → local protection → cable and terminals → load → return**. Add the battery and charger paths where provided. Identify which supplies and indications are shared by several items.

A signal transformer can have input voltage while its lamp circuit remains incomplete. Conversely, low voltage at a load can result from cable or terminal resistance rather than the power unit. The cables chapter explains that distinction further.

## 15.12 Alternative supplies and changeover

A second incoming supply and a local standby generator provide different forms of resilience. Two incoming feeders can still share an upstream dependency. A generator has its own dependencies, including starting energy, fuel, control equipment and the changeover path. Counting the available sources is therefore only the start of understanding supply security.

**Changeover equipment** transfers the load between the intended sources. The interval before the alternative supply is established matters: a generator cannot be assumed to deliver immediately after the normal supply disappears. A battery-backed circuit can bridge an interruption only within its designed capability. Follow which loads actually receive that support, rather than assuming that every load at a site is battery backed.

Supply indications must also be interpreted separately. An indication that a feeder is available is different from evidence that the load has transferred to it and is receiving the correct voltage. Source condition, changeover position and load condition answer different questions.

## 15.13 Distribution arrangements

A **radial or linear feeder** supplies successive locations along a path. A fault or isolation near its source can affect several downstream locations. A **spur** branches from a main distribution path to serve a particular load or group. The location of the branch and its protection determine how much equipment shares a dependency.

A **ring arrangement** offers paths around a loop, subject to its designed switching and operating state. An open conductor can be masked by the other path, while equipment believed disconnected from one direction may still have another feed. Recognising a loop on a drawing does not establish where it is open, how it is protected or which sources can energise it.

The supply diagram should therefore be read as a network. Mark the sources, changeover devices, section boundaries, protection, conversion equipment and loads. This reveals both the extent of a supply failure and the possible sources remaining during an isolation.

## 15.14 Fuses and discrimination

A fuse is designed to interrupt excessive current according to its characteristic. Its ampere rating is not an exact instantaneous operating threshold. The magnitude and duration of fault current, fuse type and prospective fault conditions affect its operation.

**Discrimination** means coordinating protective devices so that the device intended to clear a fault operates while supplies to unaffected circuits remain available. For example, a fault beyond a local branch fuse should, within the designed conditions, be cleared there rather than remove a whole feeder through its upstream protection. Merely fitting progressively larger ampere ratings towards the source does not establish that coordination; operating characteristics matter too.

An incorrect replacement can alter both protection and discrimination. The specified type and rating must be preserved. A higher rating is not a remedy for repeated operation: the fault or unsuitable operating condition still needs to be established through the authorised work arrangements.

### 15.14.1 Worked example: several locations lose supply

A local circuit fault occurs at the third location on a teaching radial feeder. The upstream feeder protection operates and all five locations lose supply. This differs from the intended selective response in which only the affected branch would be disconnected.

The observation directs attention to the fault path, the protective devices and their coordination. It does not prove that the upstream device is faulty: the actual fault current, downstream device, connections and specified characteristics all need to be understood.

## 15.15 Worked example: voltage falls during a point movement

An invented battery-fed circuit has 24 V at the battery terminals but only 20 V across the motor during a demand of 8 A. The 4 V difference occurs in the complete connecting path, giving an equivalent resistance of $4/8 = 0.5$ ohm at that moment. Its power loss is $8^2 \times 0.5 = 32$ W.

The arithmetic does not identify the faulty connection. It shows why the battery terminal reading alone cannot establish the motor's supply conditions. If the battery terminals also fell markedly during demand, battery internal resistance or charging condition would be additional possibilities. The complete set of measurements and circuit conditions would be needed.

## 15.16 Equipment condition and practical boundaries

Study the significance of damaged containers, leakage, corrosion, loose or overheated terminals, electrolyte condition where accessible, and charger indications. Chemical hazards and stored electrical energy exist even when the mains input is disconnected. Batteries can deliver high fault currents at relatively low voltage.

Practical inspection, isolation, loading, electrolyte handling and charging require the relevant equipment instructions, electrical safety arrangements and competence. The examples here explain the relationships; they do not prescribe a charging current, test load or acceptance limit.

## 15.17 Key points

- A supply has to suit the load's voltage, current and duty cycle.
- Battery type, cell arrangement and charging must suit the defined signalling duty.
- Internal and connection resistance matter especially under load.
- Capacity in Ah is conditional and does not itself guarantee standby duration.
- Transformers, rectifiers, chargers and distribution each perform different functions.
- Alternative supplies depend on a working changeover path; protection coordination limits the extent of a fault.

# 16. Signalling cables

**i** Learning objectives

Describe conductors, cores, insulation and sheaths; distinguish lineside, tail, power and interlinking cables; relate cable resistance to load performance; explain continuity and insulation evidence; and recognise how cable damage can affect a signalling function.

## 16.1 A cable is part of the circuit

Cables join the elements shown on a circuit diagram. They carry controls to equipment, indications back to the interlocking or display, and power to the loads. A cable fault can remove a required function or create an unintended path between functions.

An individual cable can contain circuits with different purposes. Losing several indications together can therefore point towards a shared cable, termination or supply. It does not prove that all the associated trackside items failed independently.

## 16.2 What is inside a cable?

The **conductor** carries current and is commonly copper. A **core** is an individual insulated conductor within the cable. Its **insulation** electrically separates it from other cores and surroundings. The **sheath** protects the assembly from its environment; additional armour or other layers are included where specified.

| Part | Main concern |
|---|---|
| Conductor | Resistance, current capacity and mechanical integrity |
| Core insulation | Separation between the individual electrical paths |
| Sheath | Protection against moisture, contamination and physical damage |
| Armour, screen or other specified layer | The mechanical or electrical function defined by the cable design |

A sound-looking sheath does not prove that every conductor is intact. Conversely, visible sheath damage does not describe the full extent of the internal damage. Strand construction affects flexibility, while conductor area, insulation, installation conditions and protection all influence the permissible duty.

## 16.3 Cable families and their roles

**Lineside cables** connect the interlocking area and trackside locations, or connect locations along the railway. Multicore arrangements can carry several controls or indications together. Some cores can be designated spare for future use or approved changes.

**Tail cables** connect a local housing to equipment such as signals, points, AWS apparatus or rails. They occupy an exposed part of the installation and have to accommodate the actual route and equipment interface. Track-circuit tail length and electrical characteristics can be part of the equipment design.

**Power cables** distribute the energy required by the equipment. A signalling distribution arrangement can use 650 V a.c. with local conversion, but the actual voltage and circuit must be established from the installation. Power cables must not be assumed to have unused spare cores.

**Interlinking cables** connect racks, rooms or operating equipment. Being within a building does not remove the need to identify their cores, terminal positions and circuit functions correctly.

Cable type, core count and appearance help recognition but do not prove identity. Use the markings and records associated with the actual cable.

## 16.4 Core numbering and cable ends

Core numbers let the same conductor be identified at each termination. In a concentric multicore numbering arrangement, numbering progresses through layers outwards from the centre. Viewed into the cut face, the progression runs clockwise at the **A end** and anticlockwise at the **Z end**.

The ends present opposite views of the same laid-up cable. This convention helps interpretation, but it is not a substitute for reading the core identification. Different constructions can have different identification arrangements, so the applicable cable information must be used.

A core number, terminal number and wire-function identity are different things. Core 7 can connect terminal 12 at one location to terminal 18 at another. The record must preserve that relationship rather than silently treating all three numbers as equivalent.

## 16.5 Voltage drop in signalling circuits

A point motor can impose a much larger current demand than an indication circuit. Resistance in the outgoing core, return core, contacts and terminations can therefore have different effects on the two functions. An indication that works with little current does not prove that the complete path can support the motor demand.

For a remote relay or signal lamp, compare the supply and load conditions through the complete path. A resistive termination can reduce the equipment voltage and heat under load while leaving apparent continuity. The point-movement example applies these relationships to a signalling load.

## 16.6 Routing and the mechanical environment

Cable routes can include troughing, buried sections, ducts and undertrack crossings. Tail cables pass into equipment housings through their designed entries and glands. Protection, support and allowance for movement matter at transitions between these arrangements.

Think about the path as well as the endpoints. A cable can be affected by tamping, excavation, sharp edges, unsecured lids, water, heat or equipment movement. Mechanical work near points and rail connections can affect both the cable and the equipment's reference position.

The circuit-reading chapter explains how to follow a circuit across location boundaries. Combine that electrical route with the physical cable route when studying an installation.

## 16.7 Recognising damage

| Observation | Possible significance |
|---|---|
| Cuts, abrasion or exposed inner layers | Loss of physical protection or insulation damage |
| Crushing, flattening or sharp distortion | Damaged conductors, insulation or both |
| Discolouration, melting or scorching | Heating from an electrical fault or an external source |
| Hardening, cracking or splitting | Deterioration of the sheath or insulation |
| Oil, chemical residue or other contamination | Possible material degradation and a handling hazard |
| Gnawing or disturbed protection | Animal damage that can extend beneath the visible surface |
| Corrosion or overheated terminals | Increased connection resistance or moisture-related damage |

Observation gives evidence; disturbing a suspect cable can change the fault or worsen its condition. Inspection and any handling must follow the work arrangements rather than treating a visible defect as permission to disconnect it.

## 16.8 Continuity, identity and insulation answer different questions

A **continuity test** examines whether the intended conductor provides an unbroken electrical path. **Identification** establishes that the path is the intended core and terminates where the records require. **Insulation resistance testing** examines unwanted leakage between conductors or from a conductor to earth.

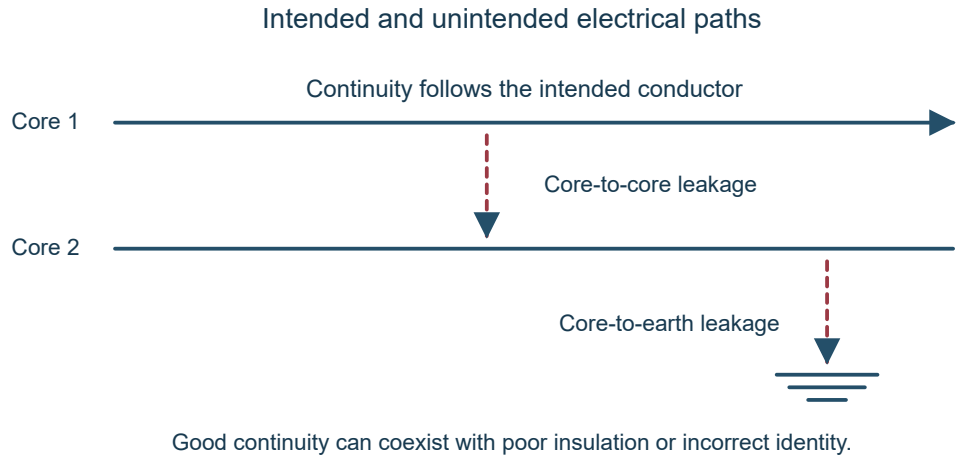



Figure 16.1: Continuity concerns the intended path; insulation concerns unwanted paths away from it.

A conductor can have good continuity and poor insulation at the same time. A cross-connected pair can also conduct perfectly while sending the function to the wrong terminal. A satisfactory result in one category cannot stand in for the others.

### 16.8.1 Understanding a loop continuity measurement

On a disconnected teaching cable, two cores linked at the remote end make a loop that can be measured from one end. Its resistance includes both conductors, the temporary link and the test connections. A good loop reading alone does not separate the resistance of each core.

A reference core can support comparison with other cores, but its own identity and integrity first have to be established. Otherwise a fault in the reference can make several healthy cores appear faulty. Parallel paths through connected equipment can also give misleading apparent continuity.

This explains the test principle. The actual connections, isolation, instruments and sequence are defined by the authorised test arrangements.

### 16.8.2 Understanding insulation evidence

Core-to-core insulation concerns separation between functions. Core-to-earth insulation concerns separation from the reference earth and surrounding conductive paths. The two results are not interchangeable.

Insulation resistance is affected by cable length, moisture, temperature and contamination, as well as the condition of the material. Compare records only when their units and relevant conditions are understood. Do not confuse a low resistance required along a conductor with the high resistance required through its insulation.

An insulation test applies its own voltage. Equipment connected to the cable can affect the result or be damaged by an unsuitable test. Instrument settings, selection of cores, treatment of connected equipment and acceptance criteria therefore come from the applicable instructions.

## 16.9 Installation, periodic checks and restoration

Initial testing establishes the required evidence for a newly installed or replaced cable. Periodic testing provides evidence about deterioration during service. A previous satisfactory result does not prove that no damage has occurred since it was recorded.

Spare cores can be included in defined monitoring arrangements, but their results cannot establish the condition of every working core. The specified maintenance regime determines which observations and tests are required.

A job is not complete merely because a test reading is satisfactory. Terminal identity, restored connections, removal and accounting of temporary links, protection and the required functional tests all belong to the complete work and testing arrangements.

## 16.10 Worked example: continuity without correct operation

A six-core teaching cable has core 4 broken and cores 5 and 6 transposed at the far end. A simple end-to-end check finds continuity on the conductors serving terminals 5 and 6, and the learner declares those connections correct.

The conclusion confuses continuity with identity. The paths conduct, but each reaches the wrong terminal. A test arrangement that proves the association between each end is needed to reveal the transposition. The break on core 4 is a separate fault; finding it does not prove that the other cores are correctly connected or adequately insulated.

## 16.11 Key points

- A cable carries defined functions through identified conductors and terminals.
- Conductor integrity and insulation integrity are different requirements.
- The complete loop determines voltage drop; a high-resistance connection can heat under load.
- Core numbering must be related to the actual termination records.
- Testing includes correct preparation and restoration, as well as obtaining readings.

# 17. Useful other sources of information

Different resources answer different questions: how equipment works, what a task requires, what applies at a location, or what a technical term means. The following are useful starting points for study and day-to-day reference.

## 17.1 OFFICIAL DOCUMENTATION

### 17.1.1 Signal maintenance specifications (SMS)

The Signal Maintenance Specifications describe the maintenance tasks and technical requirements for signalling equipment. Use the applicable specification to establish what must be inspected, maintained or measured, together with the relevant limits and recording requirements.

### 17.1.2 Signal maintenance testing handbook (SMTH)

The Signal Maintenance Testing Handbook sets out the processes, checks and tests for signalling maintenance testing within its defined scope. It helps establish the testing required after work such as equipment replacement, together with the associated planning and records.

### 17.1.3 Network Rail standards

Network Rail standards set out requirements for engineering, maintenance, safety and the management of railway assets. Use the controlled standards system to check the applicable document, issue and compliance date.

### 17.1.4 RSSB rulebook

The RSSB Rule Book contains the operational rules, modules and handbooks relevant to railway roles. For an S&T technician, it explains responsibilities and arrangements for working on signalling equipment and communicating with those controlling train movements.

### 17.1.5 Sectional Appendix

The Sectional Appendix provides route-specific information, including track layouts, line names, mileages, permissible speeds and local instructions. It is useful for understanding the railway at a particular location and relating equipment to its operational setting.

### 17.1.6 Hazard Directory

The National Hazard Directory records known hazards and location information to support planning and working on Network Rail infrastructure. Consult the relevant entries alongside the site-specific information and arrangements for the work.

### 17.1.7 Course notes

Course notes provide explanations, diagrams and worked examples from formal training, making them useful for revision and refreshing equipment knowledge. Check any task requirements or numerical limits against the applicable controlled documentation.

### 17.1.8 Network Rail e-learning

Network Rail e-learning provides online learning modules and assessments to support training, briefings and revision. Use the modules relevant to your role to prepare for courses or revisit topics that need further study.

### 17.1.9 eB suite

The eB suite is used to store, manage and retrieve engineering documents and records, including drawings. Check the document identity, revision and status to establish that the record is appropriate for the equipment and work being considered.

## 17.2 ONLINE RESOURCES

### 17.2.1 railwaycodes.org

Railway Codes is a site which lists all codes used on the railway. It's useful for looking up ELRs.

### 17.2.2 dickthesignals

dickthesignals contains a collection of signalling drawings, diagrams, manuals and technical documents that can help explain equipment and principles. Use it for background study, checking practical requirements against controlled documentation.

### 17.2.3 Glasgow S&T SharePoint

Use the Glasgow S&T SharePoint site to find local team information and shared reference material. Check the status and revision of any document before using it for a task.

## 17.3 PHYSICAL RESOURCES

### 17.3.1 Ellis' engineering encyclopaedia

*Ellis' British Railway Engineering Encyclopaedia* explains railway engineering terms, abbreviations, acronyms and jargon across several disciplines. It is a useful desk reference when an unfamiliar word appears in a drawing, document or conversation.

# Appendices

# A. Glossary

| Term | Meaning in this course |
|---|---|
| Approach control | Control of signal clearance using specified approach conditions, for example to support a speed reduction before a divergence |
| Approach locking | Locking that preserves route protection after a signal has cleared and is then replaced, until the specified release conditions are met |
| Aspect | The indication displayed by a signal |
| Axle counter | A system that determines section occupation using axle passage at defined detection points and a valid evaluator state |
| Bi-directional line | A line signalled for movements in both directions |
| Ballast resistance | The electrical resistance associated with leakage between rails through the track support and surroundings |
| Bearer | The transmission path or service carrying a communications channel |
| Bond | An electrical connection providing the required continuity between rails or around a joint |
| Call-on | A movement into an occupied line under specified signalling conditions |
| Cess | The area alongside the outside of the running lines |
| Clear indication | An indication that a section meets the system's clear-state conditions |
| Control table | A record of the functional requirements for a signalling route, point or other item |
| Crossover | Two point ends arranged to connect adjacent tracks |
| Dead locking | Point locking associated with occupation of the relevant train-detection section |
| Detection | Evidence that the specified equipment conditions have been established |
| Diamond crossing | A same-level crossing of two tracks, without necessarily providing a connection between them |
| DLM | Data Link Module, an interface to an SSI data-link path |
| ELR | Engineers' Line Reference, a code identifying a railway line or section for location records |

| Term | Meaning in this course |
|---|---|
| Facing movement | A movement approaching point toes and the divergence |
| Fail-safe | Designed so that specified failures do not produce an unsafe permission |
| FDM | Frequency Division Multiplex, sharing a path using distinct frequency channels |
| Flank protection | Protection of a route from an unwanted movement approaching from outside its direct line of travel |
| Four-foot | Conventional name for the space between a track's running rails |
| Front contact | A relay contact closed in its operated state in the basic relay description |
| Back contact | A relay contact closed in its released state in the basic relay description |
| Gauge | In the trackwork chapter, the specified distance between the inner faces of the running rails |
| Ground frame | Local controls for a limited group of equipment, with a release and protective arrangements where required |
| Headway | Separation of successive trains, expressed as distance or time under stated conditions |
| Insulated rail joint | A rail joint designed to provide electrical separation |
| Interlocking | Equipment and logic enforcing the specified relationships between routes, points, signals and other conditions |
| LDT | Long Distance Terminal, adapting an SSI link to an appropriate telecommunications bearer |
| Location case | A trackside housing for signalling interfaces, terminals, supplies and other local apparatus |
| MPM | Multiprocessor Module in the SSI architecture described |
| Normal/reverse | The two designated point positions recorded in the scheme information |
| NX | Entrance-exit route setting |
| Overlap | A defined protected distance or arrangement beyond a stop signal |
| PCM | Pulse Code Modulation, used for digital voice transmission |
| PPM | Panel Processor Module, interfacing SSI with the signaller's control equipment |
| Right-side failure | A failure producing a more restrictive result than the traffic conditions require |

| Term | Meaning in this course |
|---|---|
| Route locking | Locking that preserves the required route and protection until its release conditions are met |
| Sectional release | Progressive release of route locking behind a train under the specified sequence |
| Shunt | In train detection, a conductive connection between rails; in operations, a category of ancillary train movement |
| Six-foot | Conventional name for the space between adjacent tracks |
| SSI | In this book, the British Rail-developed Solid State Interlocking system |
| Stick circuit | A relay holding circuit using a contact of the operated relay to maintain its energisation |
| Stock rail | A fixed rail against which a switch blade closes |
| Subsidiary signal | A signal associated with a main signal and used for a specified movement while the main aspect remains at danger |
| Switch blade | A moving rail that selects the path through a set of points |
| Tail cable | A local cable between an equipment housing and a trackside item or rail connection |
| TDM | Time Division Multiplex, sharing a path using a timed sequence of channels |
| Ten-foot | Conventional name for the wider space between pairs of tracks in a four-track arrangement |
| TFM | Trackside Functional Module, an SSI interface to local inputs and outputs |
| TORR | Train Operated Route Release, using a defined train-passage sequence |
| Track circuit | Train detection using rails as part of an electrical circuit |
| Trailing movement | A movement approaching points from a converging track towards the common end |
| Up/Down | Established railway direction designations, identified from the applicable local records |
| Wrong-side failure | A failure producing a less restrictive result than the conditions justify |

## A.1 Further electrical and equipment terms

| Term | Meaning in this course |
|---|---|
| ACE | Axle Counter Evaluator in the AZLM system |
| Ampere-hour (Ah) | A unit of electric charge used for battery capacity under specified discharge conditions |

| Term | Meaning in this course |
|---|---|
| AZLM | A multi-section digital axle-counter system |
| Back drive | A mechanism providing controlled movement farther along a switch rail |
| Ballast leakage | Current taking a path between rails through their support and surroundings |
| Bonding diagram | A drawing showing individual rail connections, insulation and bonding arrangements |
| Core | An individual insulated conductor within a cable |
| Drop-shunt value | The specified shunt resistance that just causes the occupied response under stated test conditions |
| EAK | Trackside electronic unit associated with an axle-counter detection point |
| Earth-free circuit | A circuit arranged without an intentional connection of its working conductors to earth |
| Electrolyte | The ion-conducting medium in an electrical cell |
| Escapement | A mechanism coordinating phases of movement, such as unlocking, driving and relocking points |
| Float charging | Maintaining a rechargeable battery at its specified standby charging condition while supporting the load |
| Internal resistance | A source characteristic contributing to voltage fall under signalling load |
| OSS | Overspeed sensor system within TPWS |
| Phantom aspect | Apparent signal illumination caused by extraneous light rather than the intended display |
| Rectifier | A device or circuit converting alternating current into unidirectional current |
| Residual pin | A non-magnetic part maintaining a small gap in a relay’s attracted magnetic path |
| Roundel | A coloured optical element moved to select the displayed aspect in a searchlight signal |
| Secondary cell | A rechargeable electrical cell |
| Sheath | The outer protective covering of a cable |
| Snubbing | Controlled electrical braking of a motor near the end of its movement in the arrangement described |
| Specific gravity | Density relative to a reference, used for electrolyte assessment where appropriate |
| TPWS | Train Protection and Warning System |
| Track-circuit interrupter | A sacrificial element that interrupts a circuit during a defined irregular wheel movement |
| Transformer | Equipment providing the designed a.c. voltage conversion and isolation in a signalling supply |

| Term | Meaning in this course |
|---|---|
| TSS | Train stop system within TPWS |